

# Population reconstruction in acute human wounds: held-out time points, donor transfer and baseline sensitivity

Zeyu Fu

## Abstract

A smooth trajectory fitted to cross-sectional single-cell data does not establish prediction of an individual’s wound course. We tested whether population fields improve on no motion when a target time point or donor is excluded from fitting. Three healthy donors contributed intact skin and acute wounds at days 1, 7 and 30. A frozen expression representation, training-only standardization and donor-paired endpoint sampling supported separate timepoint-holdout and donor-transfer protocols. We compared four time coordinates, shared and donor-conditioned flow matching, three non-neural predictors, two training-donor counts and three training seeds. Three mouse model arms supplied a separate computational audit. Pooled Wound7 reconstruction reduced energy distance relative to stay-still by +49.9%. Excluding each donor preserved the result for the retained donors in 3/3 sensitivity runs; genuine held-out-donor transfer separately improved by a mean +75.9% in 3/3 donors. Linear days failed at the target (−18.2%), whereas rank, square-root and log-day coordinates improved. All five predictive methods met the specified split-half-reference criterion in 3/3 donor folds. Within the donor-transfer benchmark, shared flow had the smallest mean error, 0.395; conditioning improved all six training-donor comparisons without improving the overall held-out mean. Adding a second training donor reduced seed-0 mean error from 0.5496 to 0.3854. Aggregate reductions were 0.156–0.176 across seeds, with donor-block interval [0.081, 0.315], but some individual configurations varied by 0.368. In additional global day-7 and day-1 holdouts, centroid translation had lower mean donor error than shared flow (0.667 versus 0.811 and 0.929 versus 1.057, respectively). Mouse-arm changes ranged from −1.5% to +50.6%, with one animal per arm and time point. These experiments support bounded reconstruction in a three-donor acute-wound pilot. The simple interpolation controls show that a neural field is not required for the two globally missing-time tasks tested here. They establish neither DFU prognosis nor a population-level method ranking, identifiable biological trajectories, or a required cohort size.

Keywords: conditional flow matching; wound healing; held-out donor; energy distance; time coordinate; donor heterogeneity.

## 1 Introduction

Wound repair involves changing populations of inflammatory, stromal and epithelial cells, together with changes in the activity of those cells [40]. Experimental studies show that fibroblast proliferation, matrix deposition and interactions with macrophages contribute to the organization of repairing skin [26, 30]. These processes motivate a temporal description of wound tissue, but they do not imply that all wounds follow a common transcriptional clock. Acute repair in healthy skin and the course of a diabetic foot ulcer differ in their clinical setting, and a model of one cannot be assumed to predict the other [1].

Single-cell sequencing provides detailed observations of these populations while destroying the cells being measured. Samples collected at successive times therefore describe distributions of cell states rather than repeated observations of individual cells. More than one dynamic process can connect the same marginal distributions, which limits what can be identified from snapshots alone [38]. Combining transcriptional profiles with experimental lineage tracing can connect state to subsequent fate and supply information absent from expression snapshots [39]. Without such information, a smooth computational trajectory should be interpreted in terms of its stated assumptions and prediction target.

Different single-cell methods use different sources of temporal information. Trajectory inference organizes cells according to a modeled state-space topology or progression [27]. RNA velocity uses spliced and unspliced transcript information to estimate local expression change, with dynamical extensions relaxing particular steady-state assumptions [4, 14]. CellRank combines local directional information with a probabilistic description of fate transitions [15]. These approaches address questions related to, but distinct from, predicting the distribution of cells recovered from a later tissue specimen. Their outputs cannot be treated as equivalent observations of longitudinal biological motion.

Distributional models instead specify how one population measure is related to another. Optimal transport uses a chosen cost and coupling to connect observations [24, 28], while neural differential equations represent a continuous evolution in state space [7, 33]. Flow matching learns a vector field from a prescribed family of paths and conditional regression targets [17, 18, 34]. Neural stochastic differential equations additionally model diffusion, with recent work evaluating such models against lineage-informed trajectories [37]. Related transport models have also been used for perturbation prediction [5]. These developments expand the modeling choices; they do not remove the need to test whether a selected model predicts observations withheld from its construction.

That test depends on what is withheld. Removing one time point from every donor asks whether the model can reconstruct an unobserved part of the temporal sequence. Removing one donor while retaining all times in the others asks whether relationships learned from those donors transfer to the excluded person. In the latter setting, the observed source distribution of the test donor can initialize prediction without supplying its target. Neither design is equivalent to randomly separating cells from the same specimens. Cross-validation for structured data needs to reflect the dependence and extrapolation relevant to the intended application [25]. The prediction task therefore has to be defined before model performance can be interpreted.

The biological object of prediction should also be made explicit. An observed fibroblast distribution is recovered from one sampled tissue region at one time; its transcriptional variation can reflect activation, composition within the compartment and technical recovery. Descriptive expression profiles help connect latent coordinates to that material without converting the coordinates into a spatial wound map. A mean translation between available samples provides a particularly direct baseline: it asks whether moving the center of the observed population is sufficient for the evaluated target, before attributing predictive value to a flexible nonlinear field. Such a baseline must receive the same permitted donor and temporal information as the model it is compared with.

Evaluation also depends on the representation and probability measure being compared. A latent expression map makes distributional calculations tractable but determines which differences contribute to distance [20]. Quality control and normalization also determine which cells and variation enter that comparison [12]. Experience with atlas integration shows that preserving biological variation and removing technical differences can be competing objectives [21]. Energy distance supplies a distributional discrepancy with established connections to kernel-based testing [29, 31], but a distance in latent coordinates is not automatically a distance in clinical severity or gene-specific function. Pooling cells weights donors according to their recovered cell counts unless weights are changed explicitly. Preprocessing, weighting and a simple unchanged-source baseline are consequently part of the prediction question rather than incidental implementation choices.

The number of independent donors constrains what repeated computational evaluation can establish. Many cells improve the empirical description of each sampled population, while leaving donor-level biological replication unchanged [41]. Training seeds, integration resolutions and cell-sampling draws examine other sources of variability. Model selection on the same small held-out set can still lead to optimistic estimates, even when the fitting procedure is reproducible [6, 36]. Small-sample validation studies also show why performance estimates require caution when the data support many analytical choices but few independent test units [35]. These considerations favor a clearly bounded reconstruction task with explicit controls over a claim of unrestricted clinical generalization.

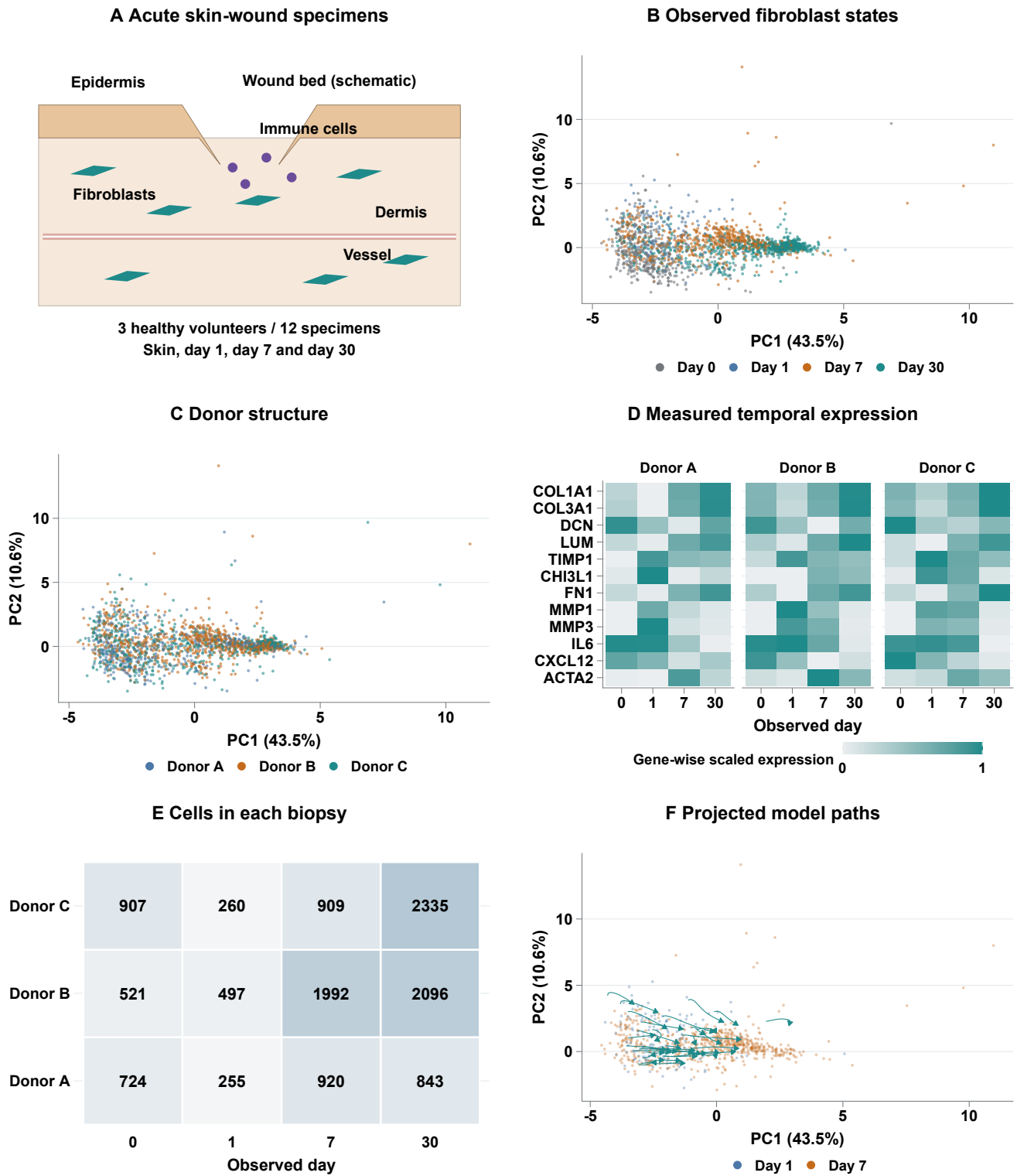
The acute human wound atlas supplies four measurement times from three healthy volunteers [19, 22]. We use this setting to ask whether moving an observed day-1 distribution under a fitted field brings it closer to held-out day-7 cells than leaving the source unchanged. Separate timepoint and donor exclusions define the two prediction tasks. Comparisons of time coordinates, shared and donor-conditioned fields, non-neural predictors, training-donor counts and seeds examine the dependence on modeling choices. Fixed-field integration, evaluation sampling and donor weighting address additional computational and measurement choices. Mouse time courses provide a separate comparison under different biological and replication conditions. The aim is to establish what this limited design can test about distributional prediction before extending its interpretation to individual wound progression.

## 2 Results

### 2.1 Observed fibroblasts define the material available for temporal prediction

Three healthy volunteers supplied intact skin and acute-wound biopsies at days 1, 7 and 30, yielding 12 specimens and 12,259 author-annotated fibroblasts (Figure 1A). A common projection displays these observed cells by collection time (Figure 1B) and by donor (Figure 1C). Its first two principal components explain 43.5% and 10.6% of the variance in the day-7 training set. Thus, nearly half of the training variance remains outside the displayed plane; visual overlap or separation cannot replace evaluation in all 15 coordinates.

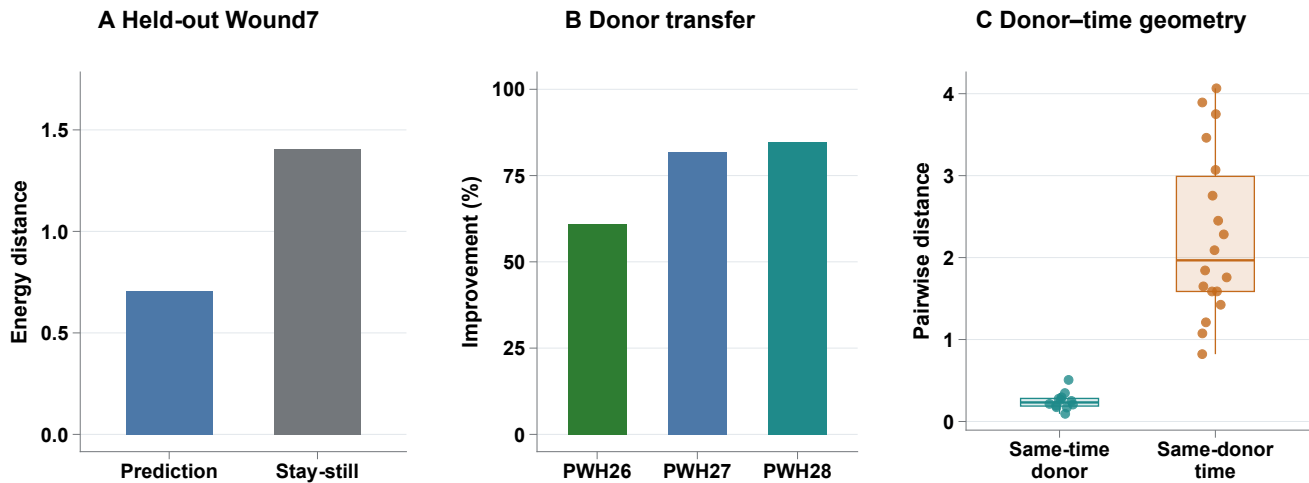
Matched raw-count profiles show the expression of matrix, inflammatory and contractile genes in each donor's four biopsies (Figure 1D). The gene-wise color scale compares a gene across samples and does not compare absolute expression between genes. Retained fibroblast counts differ across donor-time combinations (Figure 1E), so neither point density nor the number of cells provides independent temporal replication. Projected paths connect observed day-1 starting coordinates to model-predicted coordinates at the prespecified day-7 time (Figure 1F). These are numerical field trajectories; the data do not track the corresponding cells through the wound. Sampling counts are documented in Table S16.



**Figure 1.** Observed acute-wound material and its relation to model coordinates. (A) Conceptual tissue anatomy and the four biopsy times in three healthy volunteers; no histology is inferred. (B) A uniform 2,000-cell subset of 12,259 observed fibroblasts, colored by time. PCA and standardization use only day-7 training cells; axes explain 43.5% and 10.6% of training variance. (C) The identical cells colored by donor. (D) Twelve measured genes in every donor–time biopsy, using log-transformed pseudobulk counts per 10,000 and a separate zero-to-one scale for each gene. Colors do not compare absolute expression across genes. (E) Retained fibroblast counts per biopsy; cells are not independent donors. (F) Thirty-six seed-0 model paths from observed day-1 cells to the day-7 rank coordinate, over the observed day-1/day-7 display. Arrows indicate numerical motion, not tracked cell fates. All errors use the full 15-dimensional coordinates.

## 2.2 A held-out acute human time point can be reconstructed under the fixed protocol

Global timepoint holdout removes day 7 from every donor, whereas donor transfer removes one donor and retains all four times in the others. Both use the frozen representation and training-only scaling. This distinction determines whether the target time itself has been observed during training. The finalized held-out Wound7 analysis improved on the stay-still baseline by +49.9% (Figure 2A). Dropping one donor at a time preserved improvement in 3/3 retained-donor sensitivity runs; these runs predict the retained donors, not the omitted donor. The held-out donor transfer analysis also passed in 3/3 folds, with a mean improvement of +75.9% (Figure 2B). In the geometry audit, the median same-donor temporal distance was 1.967 and the median same-time donor distance was 0.233, giving a time-to-donor ratio of 8.46 (bootstrap 95% interval 6.02–13.11; Figure 2C). The mean displacement cosine was +0.948, with a descriptive bootstrap interval of 0.943–0.954. These results show that the current representation and protocol contain a transferable signal for one interval in this acute human cohort, while also showing that the geometry summary is based on only three donors (Tables S2 and S9).



**Figure 2.** Global Wound7 reconstruction, donor transfer and donor-time geometry. (A) Bars show predicted and stay-still energy distances to pooled Wound7 after excluding that time globally; relative improvement is 100 times (stay-still minus predicted distance) divided by stay-still distance, giving 49.9%. (B) Each bar is a genuinely held-out donor, with all four times retained in the other two donors; the mean relative improvement is 75.9%. Bars are point estimates without confidence intervals. (C) Points show 12 between-donor same-time distances and 18 within-donor between-time distances. Boxes span the interquartile range, centre lines are medians, and whiskers extend to extreme points within 1.5 interquartile ranges. The temporal/donor median ratio is 8.46 (10,000-draw descriptive pair-distance bootstrap 95% interval 6.02–13.11). All distances use training-standardized latent coordinates. Shared donors make these pairs dependent; three healthy donors do not provide DFU prediction validation.

The result is deliberately bounded. The donors are healthy volunteers, the wounds are acute experimental wounds and the target is one observed time point. The analysis does not contain a DFU patient trajectory or a time-to-closure endpoint. “Transfer” here means transfer across the three specified acute-wound donors under a held-out protocol; it does not mean clinical generalization.

A fixed-seed refit assessed numerical resolution and evaluation sampling. Across 25, 50, 100 and 200 RK4 steps, the target energy distance remained 0.705395 and the improvement remained 49.863%; the largest discrepancy from the 200-step coordinates was below  $6 \times 10^{-7}$  in root-mean-square units (Table S13). Repeated evaluation at 250, 500 and 1,000 cells per distribution gave median improvements of 48.46%, 48.39% and 49.26%, respectively. Gain exceeded the same-count split-half reference in all 50 draws at each count (Table S14). These results assess one fitted field and the observed cells; the ranges across draws quantify evaluation sensitivity, not additional donor replication or confidence in clinical generalization.

Using all cells and the same fixed 50-step prediction, equal cell weights gave a 49.40% improvement and equal donor mass gave 49.69%. The three donor-specific improvements under this pooled field were 47.15%, 46.92% and 40.07% (Table S15). This sensitivity changes the evaluation measure without refitting the field; it is distinct from the donor-transfer experiment and from the earlier capped-distance estimate.

## 2.3 Simple interpolation is competitive when an entire time point is missing

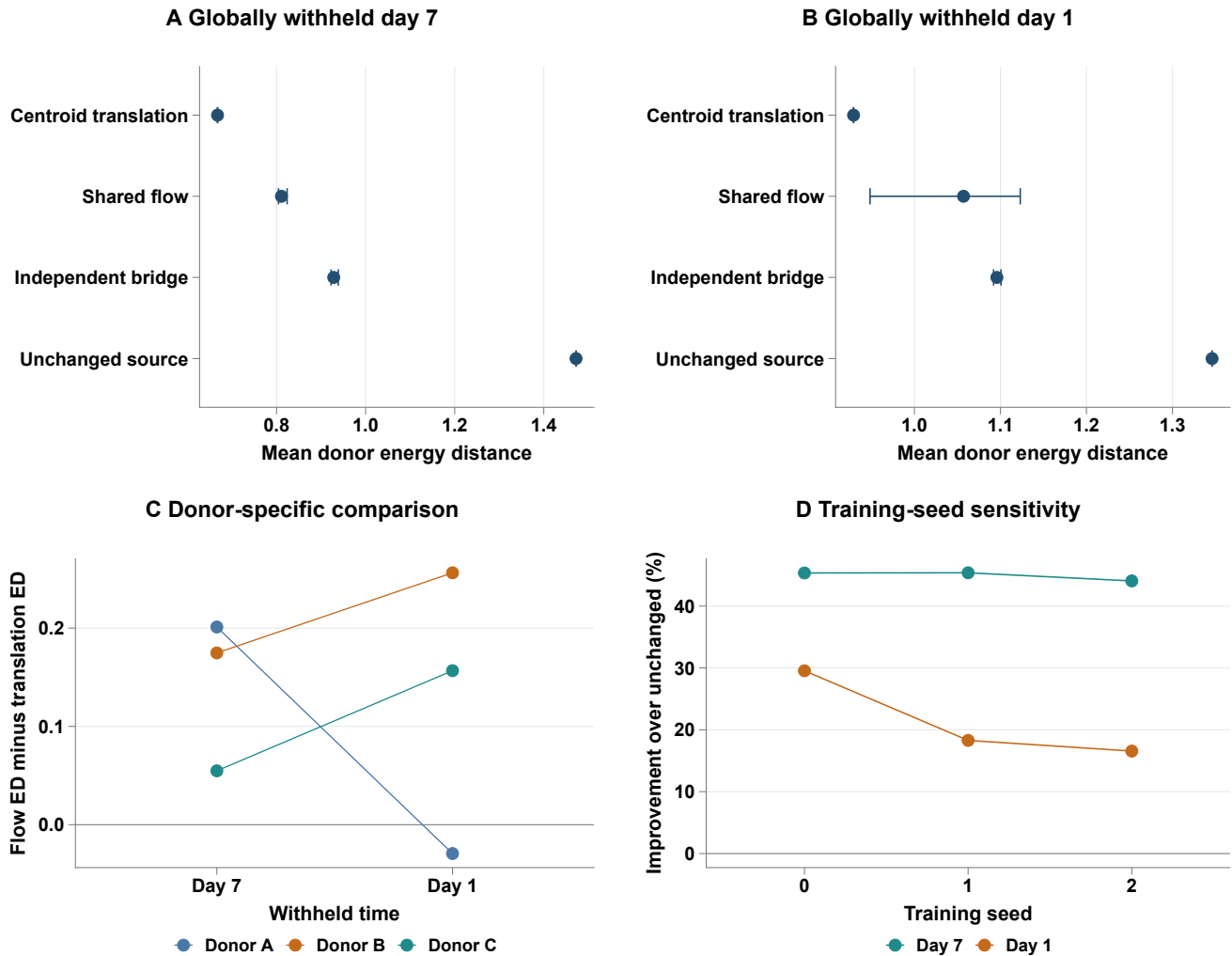
A retrospective extension tested whether improving over unchanged source requires a neural field. The additional comparisons hold out either day 7 or day 1 globally, keeping the frozen encoder, rank clock, training-only scaling and optimization budget fixed. For each task, the observed time immediately before the target supplies the source and the next observed time supplies an anchor. The source and anchor belong to the same donor; the held-time cells enter only the evaluation. Error is calculated using every available cell in all 15 coordinates within each donor, then averaged with equal donor weight. This mean of three donor-specific distances differs from the earlier pooled and capped estimands.

For withheld day 7, unchanged source had mean donor error 1.4725. Across three seeds, shared flow reduced it to 0.8110, while within-donor centroid translation reduced it further to 0.6674. An independent-endpoint linear bridge gave 0.9284 (Figure 3A). The corresponding improvements were 44.9%, 54.7% and 36.9%. Withholding day 1 instead trained on intact skin, day 7 and day

30 and predicted from intact skin. The unchanged-source error was 1.3460; shared flow, centroid translation and the independent bridge gave 1.0571, 0.9293 and 1.0959, corresponding to improvements of 21.5%, 31.0% and 18.6% (Figure 3B). The complete seed-level errors appear in Table S17.

The centroid translation had lower seed-averaged error in all three donors for withheld day 7 and in two of three for withheld day 1 (Figure 3C). The flow error ranged from 0.8044 to 0.8237 across day-7 fits, and from 0.9484 to 1.1231 across day-1 fits (Figure 3D). These ranges characterize training variation in the existing cohort; they do not provide six independent biological replications. All donor-specific comparisons are retained in Table S18.

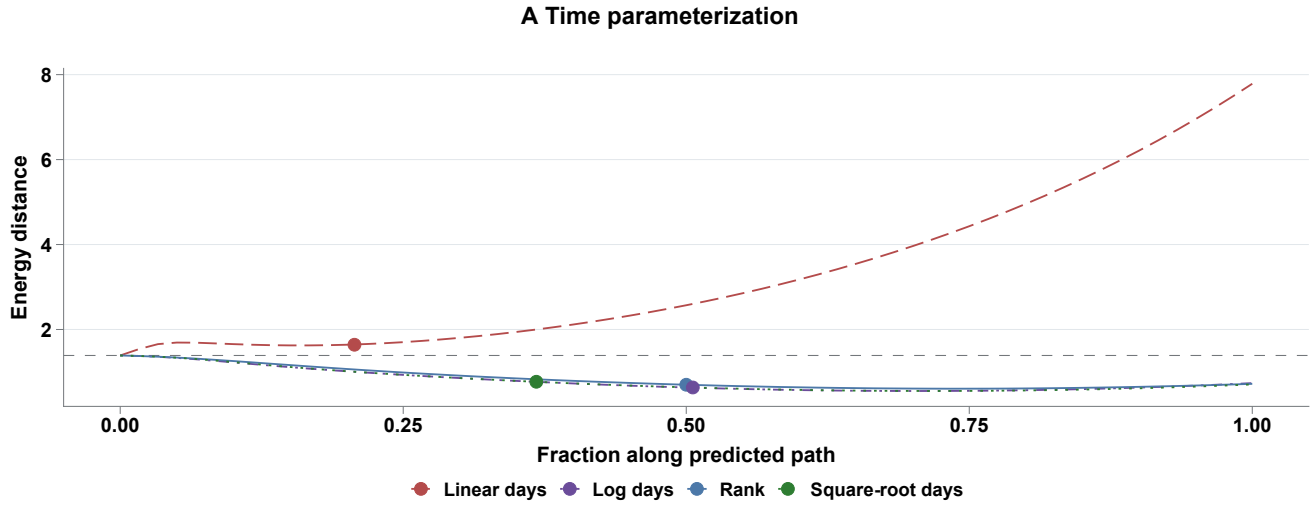
The simple predictor uses an observed later biopsy from the donor being reconstructed, which is permitted by the global timepoint-holdout task. It would not be available in prospective source-only prediction for a new donor. Conversely, the donor-transfer benchmark observes the target time in the training donors and asks a different question. Its favorable shared-field ranking cannot be generalized to a globally unobserved time. The added comparison supports temporal information in the measured distributions, while providing no evidence that a neural field is necessary for the two interpolation tasks tested here.



**Figure 3.** Simple baselines for globally missing times. All methods use the same frozen representation and training exclusions; error averages full-cell empirical energy distances within the three donors. (A) Withheld day 7, predicted from day 1 with day 30 as the observed anchor. (B) Withheld day 1, predicted from intact skin with day 7 as anchor. Points show the three-seed mean error; horizontal ranges show seed minima and maxima, not confidence intervals. Both tasks use the fixed rank-time interpolation fraction one half. (C) Donor-specific seed-averaged field error minus centroid-translation error; positive values favor translation. The later anchor belongs to the same donor, so this is retrospective interpolation rather than source-only donor transfer. (D) Each field seed's improvement over the corresponding unchanged source. Seeds and repeated distance evaluations provide no additional biological replication.

## 2.4 Time parameterization changes the held-out result

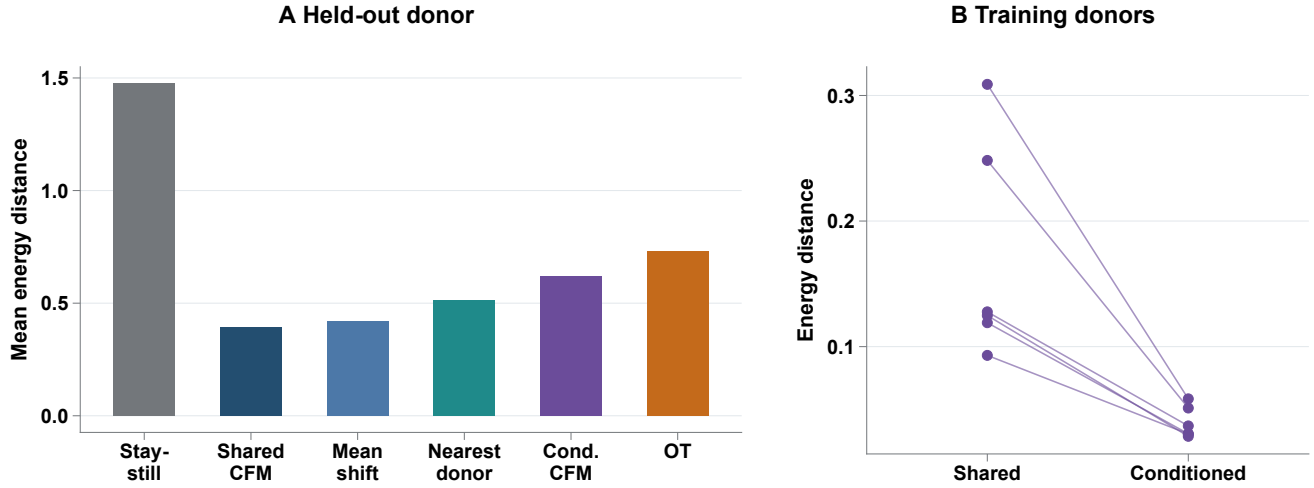
The linear days axis failed at the held-out Wound7 target, with an improvement of  $-18.2\%$  relative to stay-still. Rank, square-root days and log days met the specified noise-reference criterion, and none of the four tested axes diverged. Changing the time coordinate changes both the path specification and the optimization problem; this comparison does not isolate interval compression as a cause of failure. It does not establish log days as a universal biological clock; the transform must be prespecified and tested in a design with more independent time series (Table S3; Figure 4A).



**Figure 4.** Time-coordinate sensitivity under global Wound7 holdout in three donors. (A) Curves show energy distance along the Wound1-to-Wound30 predicted segment for rank, linear-day, square-root-day and log-day coordinates. The horizontal axis runs from the Wound1 source (0) to the Wound30 coordinate (1); line styles and colours identify the transforms. Dots are placed at each nominal Wound7 fraction with the distance evaluated at the nearest of 61 stored grid positions from 60 RK4 steps. The grey dashed line is the stay-still distance, 1.3879. Linear days gives  $-18.2\%$  relative improvement at the target; the other axes have gains above the single split-half reference, 0.0061. The full path is a retrospective diagnostic, and its minimum is not substituted for the Wound7 result. No tested axis diverges; curves do not represent confidence bands.

## 2.5 The shared field has the lowest observed held-donor error

Under the common held-donor protocol, shared flow matching had mean energy distance 0.395 and passed in 3/3 folds (Figure 5A). The non-neural mean-displacement, nearest-donor and optimal-transport baselines also passed in 3/3 folds. The donor-conditioned model passed in 3/3 held-donor folds but had a higher mean energy distance than the shared field in this comparison. In the six training-donor comparisons, conditioning improved on the shared field in 6/6 cases (Figure 5B). This difference between training-donor and held-donor is the central safeguard against interpreting capacity as generalization (Tables S4, S5 and S10).



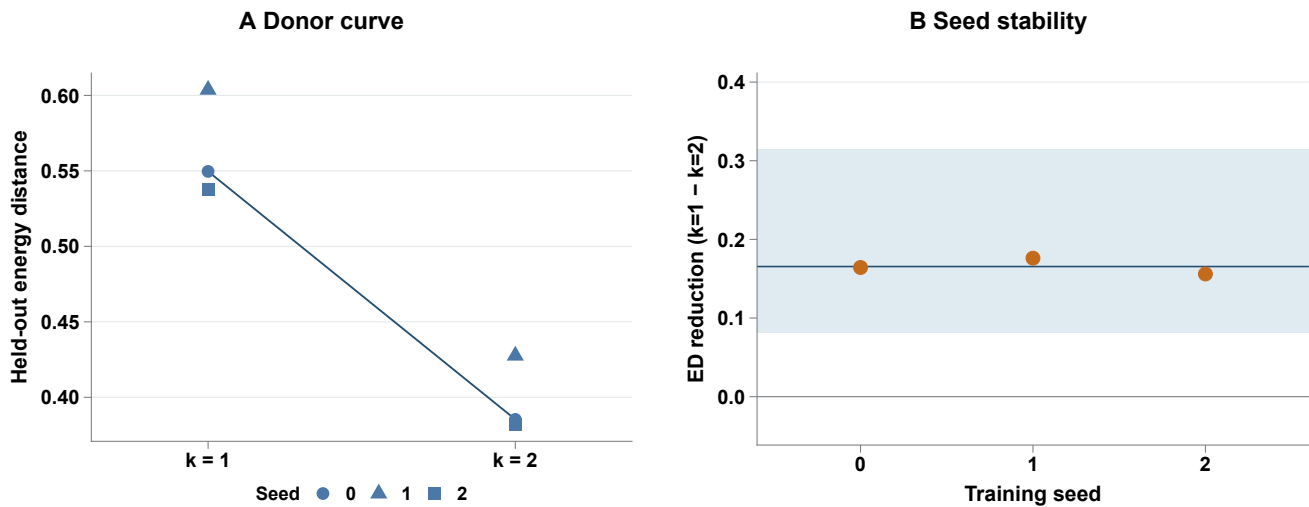
**Figure 5.** Method comparison across three held-out donors. (A) Bars are arithmetic mean energy distances for stay-still and five predictors evaluated on the same donor folds. CFM means conditional flow matching; OT is the entropic optimal-transport displacement extension. Shared CFM has the lowest point estimate (0.395). Each predictive method improves over stay-still by more than the single split-half target reference in 3/3 folds; this is a descriptive criterion, not a significance test or population method ranking. (B) Each line connects shared and donor-conditioned errors for one of six training-donor comparisons (two retained donors in each of three fits). Conditioning improves all six. These training-donor fits share donors and do not measure transfer to an unseen donor. Neither panel shows uncertainty intervals.

The data therefore do not justify selecting the conditioned field as the final model class. They support a more limited statement: the conditioned model can fit known donor contexts, while the shared model had the lowest point-estimate held-donor error in the present three-donor benchmark. Neither result identifies the model class that will work in DFU patients.

## 2.6 A second donor improves the two measured points, but $k=1$ already passes

With one training donor, mean held-out energy distance was 0.5496. With two training donors, it was 0.3854 (Figure 6A). The reduction was positive in each of the three training seeds: 0.164 at seed 0, 0.176 at seed 1 and 0.156 at seed 2. The mean-displacement control also improved at seed 0, from 0.5325 with one donor to 0.4186 with two, so the benefit of a second donor

was not unique to the neural field. The mean within-configuration training SD of 0.0717 was 3.13 times the mean split-half reference of 0.0229, showing that the reference omits a material source of run variation (Tables S11 and S12). This makes the aggregate direction reproducible across seeds (Figure 6B).



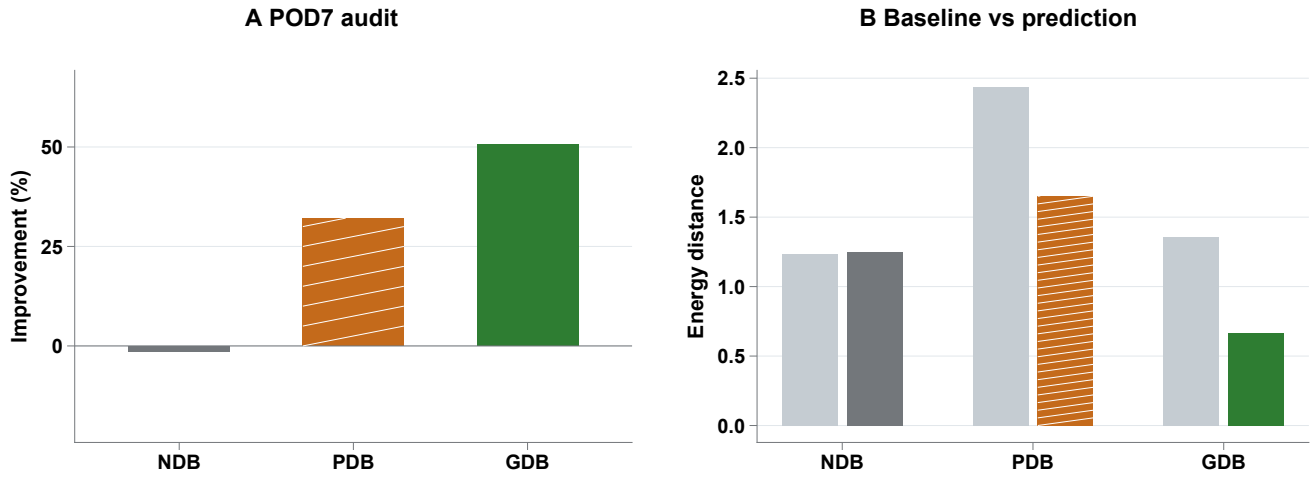
**Figure 6.** Donor-count comparison and training-seed variation. (A)  $k$  is the number of training donors. The line joins seed-0 mean held-out energy distances, 0.5496 at  $k=1$  and 0.3854 at  $k=2$ ; symbols identify seeds 0, 1 and 2. At  $k=1$ , the two single-training-donor choices are averaged within each held donor before equally weighting the three held donors. (B) Points are seed-specific reductions in mean energy distance (ED),  $k=1$  minus  $k=2$ . The horizontal line is the three-seed mean, 0.165. Shading is a percentile 95% interval [0.081, 0.315] from 4,000 resamples of three held-donor blocks after seed averaging; it is not an interval for each seed point. Evaluation subsampling is fixed across seeds. The largest configuration-specific range is 0.368. Overlapping training sets limit interval interpretation, and two  $k$  values cannot identify a plateau or required cohort size.

The same analysis shows why the curve cannot be overinterpreted. The paired donor-block reduction averaged 0.165, with a three-block bootstrap 95% interval of [0.081, 0.315]; all three held-out donor blocks improved (Table S6). The largest range for a single donor and training-subset configuration across training seeds was 0.368, while the mean within-configuration standard deviation was 0.0717. The run-to-run variation is therefore large relative to some individual improvements. More importantly,  $k=1$  already met the specified energy-distance criterion in all six  $k=1$  comparisons. The number of training donors is not an all-or-none success criterion in this dataset, and the two points cannot determine a plateau or a required cohort size. They only establish that adding a second independent context improved the current aggregate error in this pilot.

## 2.7 Three mouse arms expose model-arm heterogeneity

The all-arm audit gave different results under the same protocol. NDB changed by  $-1.5\%$  relative to stay-still and did not beat the baseline. PDB improved by  $+32.1\%$ , but its POD7 target failed an endpoint-distance ordering check, warning against a simple interpolation interpretation without proving an off-manifold trajectory. GDB improved by  $+50.6\%$  with a POD7 target satisfying that ordering check. In aggregate, 2/3 arms beat stay-still, 2/3 met the noise criterion and 2/3 satisfied the ordering check (Figure 7A). Each arm had one animal per time point, and NDB covered 59.7% of the frozen panel. These arm-level computations are useful for protocol auditing, but they do not establish a mouse biological trajectory, a cross-species equivalence or DFU transfer (Tables S7 and S8).

The paired baseline and predicted distances show the absolute error scale within each mouse arm (Figure 7B); relative improvement alone would hide these differences.



**Figure 7.** Mouse POD7 computational audit for the source-labelled NDB, PDB and GDB arms. POD7 is excluded from fitting and scaling; rank-time fields trained at POD0, POD2 and POD30 predict from the observed POD2 source. (A) Relative energy-distance improvements over stay-still are  $-1.5\%$ ,  $+32.1\%$  and  $+50.6\%$ . Hatching marks PDB, whose POD7 target fails the endpoint-distance ordering diagnostic. (B) Within each arm, the left grey bar is stay-still and the right coloured bar is prediction; hatching carries the same diagnostic meaning. The screen compares endpoint distances and does not prove or disprove geodesic interpolation. Each arm has one animal per time point and 4,182/7,002 mapped panel genes. Bars have no uncertainty intervals; cell counts do not supply independent animal replication or cross-species clinical validation.

### 3 Discussion

A fitted population field brought day-1 cells closer to the observed day-7 distribution in the acute human wound cohort. Improvement occurred when day 7 was withheld globally and under a separate protocol that withheld each donor. These comparisons support reconstruction of the measured interval in three healthy volunteers, with different information available to the model in each task. Donor transfer can use day-7 observations from other people, whereas global timepoint holdout cannot. Keeping these tasks distinct is necessary when deciding what a successful prediction says about a future dataset or an individual patient.

The global missing-time comparison changes the interpretation of model complexity. A within-donor mean translation had lower average error than the shared field for both withheld times, even though the field improved over an unchanged source. This shows why no-motion is an incomplete baseline for retrospective interpolation. The translation preserves the shape of the starting distribution and uses the observed later biopsy to estimate a displacement; its performance suggests that this simple construction captures enough of the measured change to be competitive under the chosen error metric. It does not identify the biological process producing the shift, nor imply that all aspects of fibroblast heterogeneity are preserved correctly.

That comparison does not contradict the lower shared-field error in the held-donor benchmark. In global timepoint reconstruction, the later anchor belongs to the person whose missing biopsy is reconstructed, while the target time is absent from every training donor. In donor transfer, the target time is observed in other people and the excluded donor contributes only the starting sample. These information sets define distinct prediction problems. Claims about a model's advantage therefore need to name the task and its baseline rather than treating favorable performance in either setting as evidence of general superiority.

The observed-cell maps and biopsy expression profiles make the population being transported more concrete, but they do not establish a lineage or wound mechanism. A two-dimensional projection omits variation even when fitted solely to training cells [13]. Model curves remain one construction compatible with a specified training objective, and the data do not observe proliferation, migration, death or individual cell fates along those curves. Additional time-resolved biopsies from independent donors would test missing-time generalization; spatial measurements and lineage-informed experiments would address localization and the identity of connecting processes. A DFU cohort with authoritative clinical outcomes remains a separate requirement for prognostic claims.

The learned field is a model of distributions under a specified sampling construction. Independent endpoint cells from the same donor define linear paths and regression targets; those pairings are not observed cell lineages or an estimated minimum-cost coupling. At an ideal regression solution, probability conservation refers to that constructed path measure. It does not establish that an unobserved biological distribution follows the same path. Snapshot ambiguity and mathematical identification theory therefore remain relevant even when a held-out empirical distribution is predicted accurately [16, 38]. The result supports an empirical prediction within the evaluated protocol, while leaving the biological process connecting those observations underdetermined.

Additional biological measurements would constrain a different part of the problem. Lineage tracing relates ancestry to later cell states, whereas RNA velocity derives directional information from transcript processing under a kinetic model [4, 14, 39]. Probabilistic fate mapping and neural stochastic differential equations combine such information with further modeling assumptions [15, 37]. These methods were not evaluated here, and their published performance cannot be transferred to this wound cohort. Their relevance is that they identify possible sources of information beyond marginal expression distributions. Incorpor-

rating them would require compatible assays and a new validation design, rather than only changing the function used to fit the current endpoints.

Normalization to a probability distribution introduces another biological limit. Wound repair involves proliferation, loss, recruitment and matrix remodeling, with experimental evidence for coordinated fibroblast and macrophage responses [26, 30]. The present normalized distributions do not measure absolute tissue cell counts or distinguish those processes. A mass-preserving field can describe a change in a recovered state distribution without identifying whether it arose through movement of surviving cells, altered population growth or differential recovery during sampling. This explains why a computationally coherent flow can remain incomplete as a mechanism of tissue repair.

The time-axis comparison identifies a consequential modeling choice. Changing the coordinate changes the specified bridges, their target velocities and the optimization weights. It is not simply a relabeling of time along one fitted field. Poor prediction under linear days and improvement under several alternatives show sensitivity within this experiment, but do not isolate interval compression as the cause or identify a universal biological clock. Selecting the transformation after inspecting the same target would make that target part of model selection. A later evaluation should separate selection from assessment, following the distinction emphasized in cross-validation studies [6, 36]. The scientific question is whether a fixed time rule transfers to independent observations, rather than whether one of several rules can fit the present interval.

Donor conditioning improved the fit to known donor contexts without improving the overall held-out mean relative to the shared field. Additional context can increase fitting capacity while also creating an extrapolation problem for an unfamiliar donor. A shared field already assigns different velocities at different latent positions, so different donor-average displacements do not establish that conditioning is necessary. The non-neural predictors also improved over unchanged source, showing that simple displacement rules capture part of the available structure. This makes them useful reference models when evaluating more flexible transport or response models [5]. With three test donors, the observed method ordering remains a description of this benchmark rather than an estimate of a stable population ranking.

The distance used for evaluation also determines the interpretation of improvement. Energy distance compares distributions through pairwise geometry and has a formal relationship to kernel discrepancies [29, 31]. Its value here depends on the frozen expression coordinates and training-only scaling. A lower aggregate discrepancy can coexist with errors in a rare subpopulation or in an individual gene that contributes little to that geometry. Representation benchmarking has shown why apparently successful integration must also be assessed for preservation of relevant biological variation [21]. The present comparison does not establish equivalent performance in raw gene space, spatial tissue structure or clinical outcomes. Those would be separate evaluation targets with their own measurement requirements.

The fitted and evaluated donor measures also differ. Cycling through donor-segment pairs approximately balances training contributions, whereas pooled evaluation weights cells according to yield. Equal-donor-mass evaluation of the same saved prediction preserved the improvement, and each donor separately improved under that pooled time-holdout field. This narrows the explanation that the favorable pooled result is driven only by unequal cell counts. It leaves other weighting choices unchanged, including scaling and fitting, and supplies no additional donor-transfer fold. The appropriate independent unit remains the person, not the recovered cell or the number of times a prediction is reevaluated [41].

Numerical integration and evaluation sampling were stable over the tested settings for one fixed field. Increasing Runge-Kutta resolution changed the coordinates negligibly, and repeated subsampling preserved improvement. These checks address numerical approximation and finite-cell evaluation, while training seeds revealed larger changes in some individual configurations. The aggregate benefit from a second training donor persisted across seeds, but two training sizes cannot identify a learning-curve plateau or a sufficient cohort size. Overlapping training sets correlate the held-out comparisons, and three donor blocks provide little information about the tails of a population error distribution [3, 35]. Repeated computation therefore improves characterization of this experiment without replacing biological replication.

The mouse comparison adds a useful change of setting but also exposes the limits of extrapolation. Its model arms produced different errors, each arm had one animal at each time, and the transferred gene panel had incomplete coverage. Between-time differences are consequently entangled with between-animal differences, while representation transfer adds another source of uncertainty. Acute human wounds and experimental mouse wounds do not provide independently observed diabetic-foot-ulcer courses. The next biological evaluation should preserve the intended donor and temporal exclusions under a frozen representation, clock and error measure [25]. A longitudinal clinical cohort with authoritative outcomes would be needed to determine whether distributional reconstruction contributes to staging or prognosis beyond describing the sampled acute-wound interval.

## 4 Methods

### 4.1 Data, representation and join integrity

GSE241132 contains three healthy acute-wound donors, labelled PWH26, PWH27 and PWH28, with Skin, Wound1, Wound7 and Wound30 samples [19, 22]. Among 65,521 loaded cells, 58,823 joined the author metadata, and 12,259 author-annotated fibroblasts entered the temporal analysis. The fibroblast counts were 2,152, 1,012, 3,821 and 5,274 at the four times, respectively. Annotation is joined using sample identity and original cell-row identity before quality filtering; a filtered positional row number is not a valid substitute. Earlier exploratory outputs based on positional mismatching are invalid and contribute no results here.

The expression encoder is frozen from GSE165816 [32]; the human wound cohort was not used to refit it. It maps panel-normalized counts into a  $q = 15$  dimensional Gaussian-encoder mean  $\mu$ . Its training objective combines cross-entropy on cell proportions with a standard-Gaussian KL penalty of weight 0.01; it is not a Dirichlet model or an unmodified count-likelihood objective. Only 5,383 of 7,002 panel genes are covered in the human wound input (76.9%); absent panel genes are zero-filled. The flow acts on  $\mu \in \mathbb{R}^{15}$ , not on simplex topic weights  $\theta = \text{softmax}(\mu)$ . Softmax ignores a common additive coordinate offset, whereas Euclidean distance in  $\mu$  need not. The frozen regularized encoder fixes a particular representation; the distance is not invariant to all representations producing the same simplex weights. Successful reconstruction is therefore conditional on this incomplete transferred panel and this coordinate system.

For training-cell set  $\mathcal{T}$ , scaling parameters are fitted coordinatewise on  $\mathcal{T}$  only:

$$a_j = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \mu_{ij}, \quad b_j = \left[ |\mathcal{T}|^{-1} \sum_{i \in \mathcal{T}} (\mu_{ij} - a_j)^2 \right]^{1/2}, \quad z_{ij} = \frac{\mu_{ij} - a_j}{\tilde{b}_j}. \quad (1)$$

In Equation (1),  $\tilde{b}_j = b_j$  for  $b_j > 0$  and  $\tilde{b}_j = 1$  if  $b_j = 0$ . The identical transformation is applied to source and target evaluation cells. The target is excluded from estimating  $a_j, b_j$ , and all cells of the held-out donor are excluded in donor transfer. Distances from different training folds are consequently expressed in their respective fold-standardized spaces.

## 4.2 Two distinct held-out tasks

**Timepoint reconstruction.** All Wound7 cells are withheld from fitting and scaling. Training segments are Skin→Wound1 and Wound1→Wound30 within each donor. The fitted field advances the observed Wound1 source to the specified Wound7 coordinate. The pooled analysis uses three donors. Three additional runs omit one donor and evaluate Wound7 in the remaining donors; these are sensitivity analyses, not predictions for the omitted donor.

**Donor transfer.** A donor is excluded entirely from fitting and scaling; all four time points in the other two donors are retained. Adjacent within-donor segments Skin→Wound1, Wound1→Wound7 and Wound7→Wound30 are used. The held donor's Wound1 cells initialize prediction, while their Wound7 cells are used for evaluation only. Because Wound7 is observed in training donors, this task does not test reconstruction of a globally unobserved time point. The method benchmark and donor-count experiment use this latter design. Table S1 lists the exclusions and settings.

## 4.3 Time coordinates and conditional flow matching

Let  $P_t^{(d)}$  denote the empirical distribution of standardized fibroblast coordinates for donor  $d$  at day  $t$ . Write  $s = g(t)$  for transformed time. Rank time assigns  $(0, 1/3, 2/3, 1)$  to days  $(0, 1, 7, 30)$ , and the other tested transforms are

$$g_{\text{days}}(t) = t/30, \quad g_{\text{sqrt}}(t) = \sqrt{t/30}, \quad g_{\text{log}}(t) = \log(1+t)/\log 31. \quad (2)$$

Equation (2) defines the three day-based alternatives to rank time. For one training donor and segment  $(s_a, s_b)$ , endpoints are sampled *independently* from the source and target cells of that donor. Thus the training coupling is the empirical product  $\pi_{d,ab} = P_{t_a}^{(d)} \otimes P_{t_b}^{(d)}$ . It is not an optimal-transport coupling and does not pair the same biological cell across times. With  $\lambda \sim U(0, 1)$ ,

$$z_\lambda = (1-\lambda)z_a + \lambda z_b, \quad s_\lambda = (1-\lambda)s_a + \lambda s_b, \quad u_{ab} = \frac{z_b - z_a}{s_b - s_a}. \quad (3)$$

Equation (3) defines the interpolated state, time and target velocity. Let  $\mathcal{A}$  be the list of available donor-segment pairs and  $h_{ab} = s_b - s_a > 0$ . The implementation cycles through this list, so pair counts differ by at most one batch over 8,000 steps. Its ideal uniformly weighted objective is

$$\mathcal{L}_{\text{CFM}}(\phi) = \frac{1}{|\mathcal{A}|} \sum_{(d,a,b) \in \mathcal{A}} \frac{1}{h_{ab}} \int_{s_a}^{s_b} \mathbb{E}_{\pi_{d,ab}} \left[ \frac{1}{q} \|v_\phi(z_{(s-s_a)/h_{ab}}, s) - u_{ab}\|_2^2 \right] ds. \quad (4)$$

Equation (4) averages squared error over batch examples and coordinates. The factor  $1/h_{ab}$  follows from sampling  $\lambda$  uniformly on each segment; dividing  $z_b - z_a$  by  $h_{ab}$  gives velocity per unit transformed time. Donors therefore receive approximately equal training weight within a segment, regardless of their cell yields. Scaling and the original pooled evaluation instead weight retained cells equally. These measures need not coincide. Equations (3)–(4) implement conditional flow matching with a product coupling [17, 18, 34]; they do not solve the Benamou–Brenier minimum-kinetic-energy problem [2, 24].

A pure change of time coordinate for an already specified physical-time field  $v_t$  would require

$$v_s(z, s) = \frac{v_t(z, g^{-1}(s))}{g'(g^{-1}(s))}, \quad 0 < g'(t) < \infty. \quad (5)$$

Equation (5) is the chain rule, not the procedure used in the time-axis experiment. Each tested axis instead defines new straight bridges, new segment widths and a separately fitted field with the same optimization budget. A short segment increases target-velocity magnitude and alters the loss density in time. Performance differences consequently combine modeling and optimization effects; they cannot isolate a biological clock or prove that interval compression caused an observed failure.

Following the neural ordinary-differential-equation formulation [7], predictions solve an initial-value problem and push forward the empirical source measure:

$$\frac{dz}{ds} = v_\phi(z, s), \quad z(s_a) = z_a, \quad \hat{P}_{s_b} = (\Phi_{s_a \rightarrow s_b}^\phi)_\# \hat{P}_{s_a}. \quad (6)$$

Equation (6) defines prediction from the observed source distribution. The network uses two 128-unit SiLU hidden layers and a 32-dimensional sinusoidal time embedding. Training uses Adam with learning rate  $10^{-3}$ , 8,000 steps and batches of 256. Direct target predictions use 50 fixed fourth-order Runge–Kutta steps. The time-axis diagnostic instead uses 60 steps over the full Wound1-to-Wound30 segment and reads the grid point nearest the transformed Wound7 coordinate; it does not select the grid point with the lowest target distance. A separate fixed-field resolution comparison is described below; finite-step agreement does not prove convergence to an exact solution.

#### 4.4 What distributional success does and does not identify

At the unrestricted square-integrable minimizer, the field is  $v^*(z, s) = \mathbb{E}[u_{ab} \mid z_\lambda = z, s_\lambda = s]$  under the training sampling measure. The standard squared-error decomposition leaves an irreducible conditional velocity variance plus the error in approximating  $v^*$ . Within a common observed segment, let  $P_s$  be the donor-balanced mixture of the product-coupling paths. For a smooth compactly supported test function  $f$ ,

$$\begin{aligned} \frac{d}{ds} \int f(z) dP_s(z) &= \mathbb{E}[\nabla f(z_s)^\top u_{ab}] = \int \nabla f(z)^\top v^*(z, s) dP_s(z), \\ \partial_s P_s + \nabla \cdot (P_s v^*) &= 0 \quad \text{when a density exists.} \end{aligned} \quad (7)$$

Equation (7) states probability conservation first in weak form, which does not presume that empirical atomic measures have a density. Identifying its solution with a unique deterministic ODE push-forward requires additional regularity; the weak identity alone supplies no such guarantee for empirical endpoints. Finite neural fitting and integration introduce further approximation. Moreover, Wound7 is not an endpoint in the globally withheld-time training path, and a cell-weighted source need not equal its donor-balanced source. The flow-matching identity therefore does not guarantee either the held-out biological distribution or the originally pooled evaluation score.

Normalized probability transport does not identify cell proliferation, death or ancestry. Recovery of underlying dynamics from marginal observations requires assumptions beyond matching those observations [16, 38]. The present analysis does not verify the diffusion and sampling conditions of trajectory-identification theory. With four measurement times, the values of  $g$  between observations remain analysis choices.

#### 4.5 Distance estimand, finite-sample statistic and thresholds

For independent  $X, X' \sim P$  and  $Y, Y' \sim Q$  with finite first moments, define energy distance without the optional square root [31]:

$$D_E(P, Q) = 2\mathbb{E}\|X - Y\|_2 - \mathbb{E}\|X - X'\|_2 - \mathbb{E}\|Y - Y'\|_2. \quad (8)$$

In Euclidean space, Equation (8) is nonnegative and zero exactly when  $P = Q$ . The implementation uses the V-statistic, including diagonal zero distances:

$$\begin{aligned} \widehat{D}_E(P_m, Q_n) &= \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n \|x_i - y_j\|_2 - \frac{1}{m^2} \sum_{i=1}^m \sum_{i'=1}^m \|x_i - x_{i'}\|_2 \\ &\quad - \frac{1}{n^2} \sum_{j=1}^n \sum_{j'=1}^n \|y_j - y_{j'}\|_2. \end{aligned} \quad (9)$$

Equation (9) is the exact distance between the two empirical measures, not an unbiased U-statistic for the population distance. For baseline  $B = \widehat{D}_E(P_{\text{source}}, P_{\text{target}}) > 0$  and predicted distance  $D$ , gain and relative improvement are

$$G = B - D, \quad I = G/B = 1 - D/B. \quad (10)$$

Equation (10) defines the absolute gain and the fraction reported as a percentage. Stay-still means  $P_{\text{prediction}} = P_{\text{source}}$ . Seeded subsampling caps the quadratic distance calculation at 1,500 cells per distribution for initial timepoint reconstruction and at 1,200 for donor transfer, the common method benchmark and the donor-count analyses. Counts, caps and per-fold values accompany each comparison; independently run analyses need not have identical stochastic point estimates.

The target is split once into seeded disjoint halves, defining  $N = \widehat{D}_E(P_{\text{target},1}, P_{\text{target},2})$ . The initial timepoint report asks only whether  $G > 0$ , while the time-axis, donor-transfer, method and donor-count comparisons use  $G > N$ . The latter is a sampling-scale heuristic, not a significance test with known type-I error, a repeated-resampling interval, or an error bound. Selecting the best position on a path using target distances is retrospective; the preassigned Wound7 coordinate supplies the held-out result, and the full scan is a diagnostic rather than a tuned clinical predictor.

#### 4.6 Donor context and the non-neural predictors

For a source sample, context concatenates the coordinatewise mean and population standard deviation (divisor  $n$ ),  $c_d = (\bar{z}_d, \text{sd}(z_d)) \in \mathbb{R}^{2q}$ . The conditioned field uses this context from the training segment's source. Independently for each training example  $i$ ,

$$B_i \sim \text{Bernoulli}(0.8), \quad \tilde{c}_i = B_i c_d, \quad \mathcal{L}_{\text{context}} = \mathbb{E} \left[ q^{-1} \|v_\phi(z_i, s_i, \tilde{c}_i) - u_{ab,i}\|_2^2 \right]. \quad (11)$$

Equation (11) masks the entire context vector per example, rather than masking an entire batch or individual context coordinates. At test time there is no mask, and the held donor's source alone supplies  $c_d$ . Training with zero context does not guarantee agreement with a separately trained shared field, bounded extrapolation or successful donor transfer. Both neural fields use the same donor-segment schedule and endpoint-sampling rule, although successive pseudorandom draws need not be identical. Six training-donor evaluations are reported separately from the three held-out-donor folds.

Let  $\delta_d = \bar{z}_{d,7} - \bar{z}_{d,1}$  for a training donor. Mean displacement predicts  $z + \bar{\delta}$ ; nearest-donor displacement predicts  $z + \delta_{d^*}$ , where  $d^*$  maximizes cosine similarity between the source contexts. These are translations, not fitted neural fields. The entropic transport baseline [8] uses pooled training-source and training-target cells, uniform empirical marginals  $a, b$ , and normalized squared Euclidean cost  $C$ :

$$\Gamma_\varepsilon = \arg \min_{\Gamma \geq 0, \Gamma \mathbf{1} = a, \Gamma^\top \mathbf{1} = b} \left\{ \langle C, \Gamma \rangle + \varepsilon \sum_{ij} \Gamma_{ij} (\log \Gamma_{ij} - 1) \right\}, \quad \varepsilon = 0.05. \quad (12)$$

Equation (12) defines the entropic transport plan fitted to the training endpoints. Specifically,  $C_{ij} = \|z_i - z_j\|_2^2 / \max_{kl} \|z_k - z_l\|_2^2$ , so the regularization value is relative to the largest sampled training cost. The POT implementation [10] caps Sinkhorn iteration at 2,000 steps, with at most 1,500 cells per endpoint. Row-normalized barycentres define training-source displacements; each held-out source cell receives the average displacement of its 15 nearest training-source neighbours. This is an out-of-sample displacement extension of a transport plan, not transport fitted to the held target. A finite iteration budget alone does not certify exact marginal feasibility or optimality.

#### 4.7 Donor-count estimand and dependence-aware summaries

For each held donor  $d$  and seed  $r \in \{0, 1, 2\}$ , let  $D_{1,d,r}$  average the errors from the two possible single-training-donor choices, and let  $D_{2,d,r}$  be the error with both available training donors. Equal weighting then gives

$$R_k = \frac{1}{3 \cdot 3} \sum_{d=1}^3 \sum_{r=0}^2 D_{k,d,r}, \quad \Delta_d = \frac{1}{3} \sum_{r=0}^2 (D_{1,d,r} - D_{2,d,r}), \quad \bar{\Delta} = \frac{1}{3} \sum_d \Delta_d. \quad (13)$$

Equation (13) defines the seed-averaged estimand and donor-block differences. The plotted main curve uses seed 0, with other seeds shown separately. The 4,000-draw percentile interval resamples the three  $\Delta_d$  blocks after averaging seeds [9]. Overlapping training donors still induce dependence between held-out errors [3], so even this interval is a descriptive three-block summary, not a population coverage guarantee. Seeds and alternative training subsets are not additional biological replicates. Configuration-specific ranges quantify stochastic instability. Two values of  $k$  cannot identify a plateau, a functional learning-curve model or a required number of donors.

#### 4.8 Geometry and mouse computation

Donor displacement alignment is  $\cos(\delta_d, \delta_e) = \delta_d^\top \delta_e / (\|\delta_d\| \|\delta_e\|)$ . The time-versus-donor contrast compares 18 within-donor temporal distances with 12 between-donor same-time distances. Pair-distance bootstrap intervals are descriptive because pairs share donors. The geometric screen used for endpoints  $a, b$  and held time  $h$  is

$$J = \mathbf{1} \{ \widehat{D}_E(P_a, P_b) \geq \max[\widehat{D}_E(P_a, P_h), \widehat{D}_E(P_h, P_b)] \}. \quad (14)$$

Equation (14) is a distance-ordering diagnostic. It does not imply equality in a triangle inequality, identify a geodesic, or establish that a target lies on a unique direct path. We report it as such rather than as proof of interpolability.

GSE326622 is evaluated separately for the source-labelled NDB, PDB and GDB mouse arms [23]. POD7 is excluded from fitting and scaling; POD0, POD2 and POD30 provide the training time points, with POD2 supplying the prediction source. Each mouse fit uses rank time, 6,000 optimization steps and batches of 256. The human panel's mouse-symbol mapping covers 4,182 of 7,002 genes (59.7%) in each arm, and each arm has one animal per time point. Animal identity and time are therefore confounded. The three arm computations assess protocol behaviour and representation transfer; cell numbers do not supply independent animals or cross-species clinical validation.

#### 4.9 Numerical-resolution and evaluation-sampling sensitivity

The saved human expression coordinates and annotations supply a separate seed-0 pooled fit using the stated training exclusions, 8,000 optimization steps and batch size 256. The fitted field is then fixed. Predictions integrate the same 1,012 day-1 cells with 25,

50, 100 and 200 RK4 steps to the prespecified rank-time day-7 coordinate. Distance calculations use one fixed set of 1,500 day-7 cells and double-precision pairwise distances. Root-mean-square coordinate discrepancies are measured relative to the 200-step result (Table S13).

Evaluation sensitivity fixes the 50-step prediction and varies only cell subsampling. At each count of 250, 500 and 1,000 cells per distribution, 50 seeded draws without replacement use common source and target indices for predicted and unchanged-source distances. Each draw additionally samples two disjoint target subsets of that size for the reference distance. We report median improvement, the central 95% range across draws and the count exceeding the reference (Table S14). These are conditional Monte Carlo summaries. They neither refit the field per draw nor estimate uncertainty across new donors.

#### 4.10 Evaluation weights and mathematical verification

A further analysis fixes that same 50-step field prediction, retains all 1,012 source cells and 3,821 target cells, and changes only empirical weights. For nonnegative unit-sum weights  $a, b$ , the exact weighted empirical distance is

$$\begin{aligned} \widehat{D}_{E,a,b} = & 2 \sum_{i,j} a_i b_j \|x_i - y_j\|_2 - \sum_{i,i'} a_i a_{i'} \|x_i - x_{i'}\|_2 \\ & - \sum_{j,j'} b_j b_{j'} \|y_j - y_{j'}\|_2. \end{aligned} \quad (15)$$

Equation (15) uses equal cell weights for the original pooled estimand. Equal donor mass instead assigns each cell from donor  $d$  weight  $1/(3n_d)$ , separately at source and target; predicted cells retain their source weights. We also evaluate each donor using this same pooled time-holdout field. Those rows are not new donor-transfer fits. This analysis changes neither scaling nor training weights, and it supplies no additional donor-level uncertainty estimate (Table S15).

Executable checks compare interval-scaled flow loss with an analytic translation, the context mask with a prescribed per-example draw, RK4 with translation and exponential ODE solutions, and weighted energy distance with explicitly replicated empirical samples. Invalid time spans, endpoint shapes and dropout probabilities are rejected before training. These checks verify mathematical implementation properties; they do not substitute for held-out biological evaluation. Numerical-order claims follow the usual smoothness assumptions for RK4 [11].

#### 4.11 Observed-cell geometry and raw expression context

The tissue drawing is a conceptual account of the sampled material, not a histological observation. The expression display uses a fresh raw-matrix reconstruction with author annotations joined by sample and barcode under the verified row contract. Retained fibroblast sample, donor and time ordering must match all 12,259 saved temporal coordinates. Twelve specified matrix, inflammatory and contractile genes are summarized separately in each of the 12 biopsies. Counts are pooled within the biopsy, divided by its total fibroblast count-library size, scaled to 10,000 and transformed by  $\log(1 + x)$ . Each gene is then scaled between its minimum and maximum across biopsies for display. No expression-difference test, inferred spatial position or cell lineage is obtained from this descriptive heat map.

Principal component analysis is fitted on the 8,438 standardized cells available in the global day-7 training set. Every displayed time and model prediction is projected onto that fixed basis. A seed-17 uniform subset of 2,000 observed fibroblasts supplies both time-colored and donor-colored panels. Thirty-six uniformly selected day-1 source cells supply the displayed paths, sampled at 11 evenly spaced model times under the seed-0 field. Intermediate coordinates are obtained by 50-step RK4 integration from the same source to each requested time. The projection can omit relevant state variation [13]; all prediction errors continue to use the full 15-dimensional coordinates. Arrows indicate numerical evolution under the fitted field and do not identify the fate of observed cells.

#### 4.12 Global missing-time baselines and a second interpolation target

A separate retrospective comparison fixes two global exclusions, day 7 and day 1. The original rank coordinates remain 0, 1/3, 2/3, 1 even after a time is excluded. For withheld day 7, training segments connect day 0 to day 1 and day 1 to day 30; for withheld day 1, they connect day 0 to day 7 and day 7 to day 30. Standardization excludes every held-time cell. All three donors remain in the training cohort, and the original within-donor product endpoint sampling, network, learning rate, 8,000 optimization steps and batch size 256 are retained. Three independent fits use seeds 0, 1 and 2. The extension does not retune the clock or use held-time error to choose an integration endpoint.

Let  $a < h < b$  be the source, held and next available times on this rank axis, and let  $\alpha = (h - a)/(b - a)$ . For a source cell  $z_i$  from donor  $d$ , the two explicit interpolation controls are

$$\hat{z}_i^{\text{mean}} = z_i + \alpha(\bar{z}_{d,b} - \bar{z}_{d,a}), \quad \hat{z}_i^{\text{bridge}} = (1 - \alpha)z_i + \alpha Z_{d,b}, \quad Z_{d,b} \sim \hat{P}_{d,b}. \quad (16)$$

Equation (16) translates the entire source distribution by its donor's interpolated mean displacement or independently samples an anchor cell for each source cell. Anchor draws use replacement and seeds 0, 1 and 2; the mean translation is deterministic. Both tasks have  $\alpha = 1/2$ . Source and anchor biopsies are available in training, while the target biopsy is withheld. These baselines do not solve an optimal-transport coupling. The bridge can contract or otherwise alter dispersion, whereas translation preserves the source's centered distribution. Neither construction is a measured cell trajectory.

To prevent donors with more recovered cells from dominating this comparison, compute the empirical energy distance separately for each donor using all its target cells and all source-derived predictions, then use

$$R_{m,r,h} = \frac{1}{3} \sum_{d=1}^3 \widehat{D}_E(\widehat{P}_{m,r,d,h}, \widehat{P}_{d,h}), \quad I_{m,r,h} = 1 - \frac{R_{m,r,h}}{R_{\text{unchanged},h}}. \quad (17)$$

Equation (17) defines the donor-averaged error and corresponding improvement; it is not a pooled distance with equal donor mass because it omits cross-donor distance terms. Distances use double precision in all 15 training-standardized coordinates, with no cell cap. We retain every donor and seed, average seeds only after each donor comparison, and report seed ranges without interpreting them as biological confidence intervals. The source and anchor differ between tasks, so their absolute errors are conditional on different training scalers and targets. Access to each donor’s later anchor makes this retrospective reconstruction, not prospective source-only prognosis.

## Data and reproducibility

Human acute-wound data and author cell annotations are available as GSE241132 [19, 22]. GSE165816 supplies the independently frozen expression model [32]; its DFU clinical labels are not used to train the human temporal fields. Mouse data are from GSE326622 [23]. This secondary computational study recruited no participants and performed no new animal experiments. Source-study consent and ethical approvals remain those of the original investigators. Sample identities, preprocessing exclusions, numerical settings and all reported comparisons are defined in the methods and supplementary tables. Analysis code is released under the MIT License. Manuscripts, figures and source-study data remain under their respective rights and repository terms. The accompanying software archive, authored by Zeyu Fu, is available at <https://github.com/PeterPonyu/wound-population-dynamics>. An independently archived software release is available at <https://doi.org/10.5281/zenodo.22875664>.

## References

- [1] D. G. Armstrong, A. J. Boulton, and S. A. Bus. Diabetic Foot Ulcers and Their Recurrence. *New England Journal of Medicine*, 376(24): 2367–2375, 2017. doi: 10.1056/nejmra1615439.
- [2] J.-D. Benamou and Y. Brenier. A computational fluid mechanics solution to the Monge–Kantorovich mass transfer problem. *Numerische Mathematik*, 84(3):375–393, 2000. doi: 10.1007/s002110050002.
- [3] Y. Bengio and Y. Grandvalet. No Unbiased Estimator of the Variance of K-Fold Cross-Validation. *Journal of Machine Learning Research*, 5:1089–1105, 2004. URL <https://www.jmlr.org/papers/v5/grandvalet04a.html>.
- [4] V. Bergen, M. Lange, S. Peidli, F. A. Wolf, and F. J. Theis. Generalizing RNA velocity to transient cell states through dynamical modeling. *Nature Biotechnology*, 38(12):1408–1414, 2020. doi: 10.1038/s41587-020-0591-3.
- [5] C. Bunne, S. G. Stark, G. Gut, J. S. del Castillo, M. Levesque, K.-V. Lehmann, L. Pelkmans, A. Krause, and G. Rätsch. Learning single-cell perturbation responses using neural optimal transport. *Nature Methods*, 20(11):1759–1768, 2023. doi: 10.1038/s41592-023-01969-x.
- [6] G. C. Cawley and N. L. C. Talbot. On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010. URL <https://www.jmlr.org/papers/v11/cawley10a.html>.
- [7] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud. Neural Ordinary Differential Equations. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL <https://arxiv.org/abs/1806.07366>.
- [8] M. Cuturi. Sinkhorn Distances: Lightspeed Computation of Optimal Transport. In *Advances in Neural Information Processing Systems*, volume 26, 2013. URL <https://proceedings.neurips.cc/paper/2013/hash/af21d0c97db2e27e13572cbf59eb343d-Abstract.html>.
- [9] B. Efron and R. J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall, New York, 1993. ISBN 9780412042317.
- [10] R. Flamary, N. Courty, A. Gramfort, M. Z. Alaya, A. Boisbunon, S. Chambon, L. Chapel, A. Corenflos, K. Fatras, N. Fournier, L. Gautheron, N. T. H. Gayraud, H. Janati, A. Rakotomamonjy, I. Redko, A. Rolet, A. Schutz, V. Seguy, D. J. Sutherland, R. Tavenard, A. Tong, and T. Vayer. POT: Python Optimal Transport. *Journal of Machine Learning Research*, 22(78):1–8, 2021. URL <https://www.jmlr.org/papers/v22/20-451.html>.
- [11] E. Hairer, S. P. Nørsett, and G. Wanner. *Solving Ordinary Differential Equations I: Nonstiff Problems*. Springer, 2 edition, 1993. doi: 10.1007/978-3-540-78862-1.
- [12] L. Heumos, A. C. Schaar, C. Lance, A. Litinetskaya, F. Drost, L. Zappia, M. D. Lücken, D. C. Strobl, J. Henao, F. Curion, Single-cell Best Practices Consortium, H. Aliee, M. Ansari, P. Badia-i Mompel, M. Büttner, E. Dann, D. Dimitrov, L. Dony, A. Frishberg, D. He, S. Hediye-zadeh, L. Hetzel, I. L. Ibarra, M. G. Jones, M. Lotfollahi, L. D. Martens, C. L. Müller, M. Nitzan, J. Ostner, G. Palla, R. Patro, Z. Piran, C. Ramírez-Suástegui, J. Saez-Rodriguez, H. Sarkar, B. Schubert, L. Sikkema, A. Srivastava, J. Tanevski, I. Virshup, P. Weiler, H. B. Schiller, and F. J. Theis. Best practices for single-cell analysis across modalities. *Nature Reviews Genetics*, 24(8):550–572, 2023. doi: 10.1038/s41576-023-00586-w.
- [13] H. Huang, Y. Wang, C. Rudin, and E. P. Browne. Towards a comprehensive evaluation of dimension reduction methods for transcriptomic data visualization. *Communications Biology*, 5:719, 2022. doi: 10.1038/s42003-022-03628-x.
- [14] G. La Manno, R. Soldatov, A. Zeisel, E. Braun, H. Hochgerner, V. Petukhov, K. Lidschreiber, M. E. Kastri, P. Lönnerberg, A. Furlan, J. Fan, L. E. Borm, Z. Liu, D. van Bruggen, J. Guo, X. He, R. Barker, E. Sundström, G. Castelo-Branco, P. Cramer, I. Adameyko, S. Linnarsson, and P. V. Kharchenko. RNA velocity of single cells. *Nature*, 560(7719):494–498, 2018. doi: 10.1038/s41586-018-0414-6.

- [15] M. Lange, V. Bergen, M. Klein, M. Setty, B. Reuter, M. Bakhti, H. Lickert, M. Ansari, J. Schniering, H. B. Schiller, D. Pe'er, and F. J. Theis. CellRank for directed single-cell fate mapping. *Nature Methods*, 19(2):159–170, 2022. doi: 10.1038/s41592-021-01346-6.
- [16] H. Lavenant, S. Zhang, Y.-H. Kim, and G. Schiebinger. Toward a mathematical theory of trajectory inference. *The Annals of Applied Probability*, 34(1A):428–500, 2024. doi: 10.1214/23-AAP1969.
- [17] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow Matching for Generative Modeling. *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2210.02747>.
- [18] Y. Lipman, M. Havasi, P. Holderrieth, N. Shaul, M. Le, B. Karrer, R. T. Q. Chen, D. Lopez-Paz, H. Ben-Hamu, and I. Gat. Flow Matching Guide and Code. arXiv, 2024. URL <https://arxiv.org/abs/2412.06264>.
- [19] Z. Liu, X. Bian, L. Luo, Å. K. Björklund, L. Li, L. Zhang, Y. Chen, L. Guo, J. Gao, C. Cao, J. Wang, W. He, Y. Xiao, L. Zhu, K. Annusver, N. H. Gopee, D. Basurto-Lozada, D. Horsfall, C. L. Bennett, M. Kasper, M. Haniffa, P. Sommar, D. Li, and N. X. Landén. Spatiotemporal single-cell roadmap of human skin wound healing. *Cell Stem Cell*, 32(3):479–498.e8, 2025. doi: 10.1016/j.stem.2024.11.013.
- [20] R. Lopez, J. Regier, M. B. Cole, M. I. Jordan, and N. Yosef. Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15(12):1053–1058, 2018. doi: 10.1038/s41592-018-0229-2.
- [21] M. D. Luecken, M. Büttner, K. Chaichoompu, A. Danese, M. Interlandi, M. F. Mueller, D. C. Strobl, L. Zappia, M. Dugas, M. Colomé-Tatché, and F. J. Theis. Benchmarking atlas-level data integration in single-cell genomics. *Nature Methods*, 19(1):41–50, 2022. doi: 10.1038/s41592-021-01336-8.
- [22] NCBI Gene Expression Omnibus. GSE241132: Spatiotemporal Single-Cell Roadmap of Human Skin Wound Healing. Gene Expression Omnibus accession, 2024. URL <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE241132>.
- [23] NCBI Gene Expression Omnibus. GSE326622: Mechanotransduction unifies healthy non-diabetic wound healing over time by promoting a Cd14+/Cd11a+ fibroblast subpopulation. Gene Expression Omnibus accession, 2026. URL <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE326622>.
- [24] G. Peyré and M. Cuturi. Computational Optimal Transport with Applications to Data Sciences. *Foundations and Trends in Machine Learning*, 11(5–6):355–607, 2019. doi: 10.1561/22000000073.
- [25] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Aroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, and C. F. Dormann. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017. doi: 10.1111/ecog.02881.
- [26] E. Rognoni, A. O. Pisco, T. Hiratsuka, K. H. Sipilä, J. M. Belmonte, S. A. Mobasseri, C. Philippeos, R. Dilão, and F. M. Watt. Fibroblast state switching orchestrates dermal maturation and wound healing. *Molecular Systems Biology*, 14(8):MSB178174, 2018. doi: 10.15252/msb.20178174.
- [27] W. Saelens, R. Cannoodt, H. Todorov, and Y. Saeys. A comparison of single-cell trajectory inference methods. *Nature Biotechnology*, 37(5):547–554, 2019. doi: 10.1038/s41587-019-0071-9.
- [28] G. Schiebinger, J. Shu, M. Tabaka, B. Cleary, V. Subramanian, A. Solomon, J. Gould, S. Liu, S. Lin, P. Berube, L. Lee, J. Chen, J. Brumbaugh, P. Rigollet, K. Hochedlinger, R. Jaenisch, A. Regev, and E. S. Lander. Optimal-Transport Analysis of Single-Cell Gene Expression Identifies Developmental Trajectories in Reprogramming. *Cell*, 176(4):928–943.e22, 2019. doi: 10.1016/j.cell.2019.01.006.
- [29] D. Sejdinovic, B. Sriperumbudur, A. Gretton, and K. Fukumizu. Equivalence of distance-based and RKHS-based statistics in hypothesis testing. *The Annals of Statistics*, 41(5), 2013. doi: 10.1214/13-aos1140.
- [30] B. A. Shook, R. R. Wasko, G. C. Rivera-Gonzalez, E. Salazar-Gatzimas, F. López-Giráldez, B. C. Dash, A. R. Muñoz-Rojas, K. D. Aultman, R. K. Zwick, V. Lei, J. L. Arbiser, K. Miller-Jensen, D. A. Clark, H. C. Hsia, and V. Horsley. Myofibroblast proliferation and heterogeneity are supported by macrophages during skin repair. *Science*, 362(6417):eaar2971, 2018. doi: 10.1126/science.aar2971.
- [31] G. J. Székely and M. L. Rizzo. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013. doi: 10.1016/j.jspi.2013.03.018.
- [32] G. Theodoridis, B. E. Thomas, D. Sarkar, et al. Single cell transcriptomic landscape of diabetic foot ulcers. *Nature Communications*, 13:181, 2022. doi: 10.1038/s41467-021-27801-8.
- [33] A. Tong, J. Huang, G. Wolf, D. van Dijk, and S. Krishnaswamy. TrajectoryNet: A Dynamic Optimal Transport Network for Modeling Cellular Dynamics. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 9526–9536, 2020. URL <https://proceedings.mlr.press/v119/tong20a.html>.
- [34] A. Tong, K. Fatras, N. Malkin, G. Huguet, Y. Zhang, J. Rector-Brooks, G. Wolf, and Y. Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport. *Transactions on Machine Learning Research*, 2024. URL <https://arxiv.org/abs/2302.00482>.
- [35] A. Vabalas, E. Gowen, E. Poliakoff, and A. J. Casson. Machine learning algorithm validation with a limited sample size. *PLOS ONE*, 14(11):e0224365, 2019. doi: 10.1371/journal.pone.0224365.
- [36] S. Varma and R. Simon. Bias in error estimation when using cross-validation for model selection. *BMC Bioinformatics*, 7(1):91, 2006. doi: 10.1186/1471-2105-7-91.
- [37] M. E. Vinyard, A. W. Rasmussen, R. Li, A. M. Klein, G. Getz, and L. Pinello. Learning cell dynamics with neural differential equations. *Nature Machine Intelligence*, 7(12):1969–1984, 2025. doi: 10.1038/s42256-025-01150-3.
- [38] C. Weinreb, S. Wolock, B. K. Tusi, M. Socolovsky, and A. M. Klein. Fundamental limits on dynamic inference from single-cell snapshots. *Proceedings of the National Academy of Sciences*, 115(10), 2018. doi: 10.1073/pnas.1714723115.
- [39] C. Weinreb, A. Rodriguez-Fraticelli, F. D. Camargo, and A. M. Klein. Lineage tracing on transcriptional landscapes links state to fate during differentiation. *Science*, 367(6479):eaaw3381, 2020. doi: 10.1126/science.aaw3381.
- [40] H. N. Wilkinson and M. J. Hardman. Wound healing: cellular mechanisms and pathological outcomes. *Open Biology*, 10(9):200223, 2020. doi: 10.1098/rsob.200223.

[41] K. D. Zimmerman, M. A. Espeland, and C. D. Langefeld. A practical solution to pseudoreplication bias in single-cell studies. *Nature Communications*, 12(1):738, 2021. doi: 10.1038/s41467-021-21038-1.

## Supplementary methods and numerical results

### S1 Protocol and estimator specification

Table S1 separates the missing-time and missing-donor questions. Training endpoints are paired by donor, while the individual cells drawn from those endpoints are independent. Holding out a sample in an already trained donor is not equivalent to holding out that donor. The representation is frozen before all temporal comparisons.

**Table S1.** Training and evaluation information for the six temporal protocols. Human data comprise three donors at Skin, Wound1, Wound7 and Wound30; mouse arms use postoperative days (POD) 0, 2, 7 and 30. Scaling follows the listed training exclusions. Source-only context at test time is permitted for donor-conditioned prediction, while the evaluation target is excluded from fitting in each held-out task. The explicit target column distinguishes prediction for an excluded donor from sensitivity analyses evaluated only in retained donors.

| Protocol                   | Training data                                     | Evaluation target and interpretation                                                                  |
|----------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Global Wound7 holdout      | Skin, Wound1 and Wound30 from the included donors | Their Wound7 distributions; one globally missing measurement time.                                    |
| Drop-one-donor sensitivity | Same time exclusions after removing one donor     | Wound7 in the retained donors; sensitivity to donor composition, not prediction of the omitted donor. |
| Donor transfer             | All four times in the other two donors            | Omitted donor’s Wound7 from their Wound1 source; observed interval in the training cohort.            |
| Method benchmark           | Same donor-transfer folds                         | Shared/conditioned fields and three non-neural predictors; training-donor fits shown separately.      |
| Donor-count curve          | One or two of the available training donors       | Same held donor and target; seeds and subsets do not add independent donors.                          |
| Mouse POD7 audit           | POD0, POD2 and POD30 in one source arm            | POD7 from POD2; one animal per time point confounds time and individual.                              |

### S2 Numerical prediction sequence

The reproducible evaluation order is: (1) resolve cell/sample identities and select the lineage; (2) define the held-out cells; (3) fit standardization only on training cells; (4) sample training endpoints and optimize the specified objective; (5) transport source cells with the fixed numerical integrator; (6) compute target distance, stay-still distance and the single split-half reference; (7) aggregate within the stated donor, seed and subset units. Target-distance path scans are diagnostics after prediction and are not used to select the reported target coordinate.

For completeness, the fixed-step RK4 update for step  $h$  is

$$\begin{aligned}
 k_1 &= v(z_n, s_n), & k_2 &= v(z_n + hk_1/2, s_n + h/2), \\
 k_3 &= v(z_n + hk_2/2, s_n + h/2), & k_4 &= v(z_n + hk_3, s_n + h), \\
 z_{n+1} &= z_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4), & h &= (s_b - s_a)/50.
 \end{aligned} \tag{S1}$$

Equation (S1) gives the numerical update for the direct target predictions. The conditioned field holds the source-derived context fixed throughout this integration. Under standard smoothness assumptions RK4 has fourth-order global convergence [11], but this property does not establish the actual error of the fitted field or certify the chosen step count in these experiments.

### S3 Limits of the noise and geometric checks

The finite empirical energy distance is nonzero when two independently sampled subsets represent the same population. The split-half reference therefore indicates one realized scale of target-sampling variation. It changes with cell number, latent scaling and random split, and does not include model-fit uncertainty. Comparing a prediction gain to it is a practical diagnostic with no established frequentist size in this analysis.

Likewise, the endpoint-distance ordering condition cannot show that a point lies on a line or a geodesic. In a metric space, even being closer to both endpoints than the endpoints are to each other is only a necessary-looking geometric screen, not an interpolation identity; here the unsquared-root energy quantity is used by convention and no triangle-inequality argument is invoked. PDB is consequently labelled as failing the ordering screen, not as a proven off-path biological state.

The fitted transport moves probability mass between empirical distributions; it does not model absolute cell expansion, death, migration into a biopsy or clinical closure. These unobserved processes and the non-uniqueness of couplings prevent a mechanistic interpretation of an individual streamline.

## S4 Uncertainty and numerical results

Tables S2–S9 provide target errors, time-axis comparisons, the six-entry benchmark including stay-still, donor-specific method errors, the seed-averaged donor curve, mouse-arm contrasts and descriptive geometry intervals. All five predictive methods meeting a criterion in three folds is not five independent replications, because they share test donors. The lowest observed mean is reported without a population-level method ranking.

In the donor curve, two alternative single-donor fits contribute to each  $k = 1$  donor/seed value. The three-seed mean improvement is then aggregated over three held donors. Because training sets overlap between folds, the block interval does not estimate calibrated population or predictive uncertainty. Extending the curve requires new independent donors before any plateau rule can be evaluated. There is no statistically identified minimum cohort size from the two observed  $k$  values. Table S10 reports all six training-donor comparisons. Tables S11 and S12 separate mean-displacement controls, seed-level means and configuration-specific training variability. Table S8 lists the cell counts behind each mouse computation; each cell group still represents only one animal. The retained-donor sensitivity rows in Table S2 evaluate the two donors left in the fit, not the excluded donor.

**Table S2.** Human Wound7 evaluation in GSE241132. ED denotes the empirical energy-distance V-statistic in training-standardized latent space; lower values indicate closer predicted and observed distributions. The pooled row excludes Wound7 globally. Drop-one rows exclude the named donor and evaluate pooled Wound7 only in the two retained donors. Transfer rows instead evaluate the excluded donor after fitting all four time points of the other donors. Improvement is 100 times (stay-still ED minus predicted ED) divided by stay-still ED. Split-half ED uses one seeded target split and is not a confidence limit. Every row is a point estimate; these dependent fits do not create additional donors.

| Evaluation                | Stay-still ED | Predicted ED | Split-half ED | Improvement |
|---------------------------|---------------|--------------|---------------|-------------|
| Pooled time holdout       | 1.4069        | 0.7054       | 0.0043        | 49.86%      |
| Drop PWH26; retained pair | 1.3357        | 0.7434       | 0.0043        | 44.34%      |
| Drop PWH27; retained pair | 1.2780        | 0.6258       | 0.0128        | 51.04%      |
| Drop PWH28; retained pair | 1.5460        | 0.7638       | 0.0080        | 50.59%      |
| PWH26 transfer            | 1.8221        | 0.7110       | 0.0171        | 60.98%      |
| PWH27 transfer            | 1.5355        | 0.2786       | 0.0096        | 81.85%      |
| PWH28 transfer            | 1.0704        | 0.1634       | 0.0418        | 84.73%      |

**Table S3.** Four time coordinates under global Wound7 holdout. The reference distance is 1.3879 and the split-half distance is 0.0061. ED denotes empirical energy distance in training-standardized latent space. Wound7 time is the global transformed coordinate; the reported target distance is evaluated at the nearest point of the 60-step Wound1-to-Wound30 scan. Improvement is 100 times the gain divided by stay-still distance. Rows use the same three donors, with Wound7 excluded globally; the path minimum is not substituted for target error, and no interval is shown.

| Time coordinate | Wound7 time | Target ED | Improvement | Diverged |
|-----------------|-------------|-----------|-------------|----------|
| rank            | 0.6667      | 0.6994    | 49.61%      | False    |
| days            | 0.2333      | 1.6411    | -18.24%     | False    |
| sqrt days       | 0.4830      | 0.7695    | 44.56%      | False    |
| log days        | 0.6055      | 0.6344    | 54.29%      | False    |

**Table S4.** Common Wound1-to-Wound7 benchmark across the three held-out human donors. ED is empirical energy distance in each training-standardized fold; reported means equally weight donors. Gain is stay-still ED minus method ED. Pass count records gains above the single split-half target reference, a descriptive criterion rather than a significance test. CFM denotes conditional flow matching and OT the entropic optimal-transport displacement extension. All methods share test donors; point estimates without confidence intervals do not establish a population ranking.

| Method                | Mean ED | Mean gain | Pass count |
|-----------------------|---------|-----------|------------|
| Stay-still            | 1.4760  | 0.0000    | 0/3        |
| Shared CFM            | 0.3947  | 1.0813    | 3/3        |
| Mean displacement     | 0.4186  | 1.0574    | 3/3        |
| Nearest donor         | 0.5147  | 0.9613    | 3/3        |
| Conditioned CFM       | 0.6176  | 0.8584    | 3/3        |
| Entropic OT extension | 0.7284  | 0.7476    | 3/3        |

**Table S5.** Empirical energy distances for each method and held-out human donor in the common benchmark. Each column uses the other two donors for fitting and scaling, the named donor's Wound1 cells as source and its Wound7 cells only for evaluation. Lower is better; stay-still retains the source distribution. CFM means conditional flow matching; OT is the entropic transport displacement extension. Entries are point estimates with a 1,200-cell distance cap. Training draws and this cap differ from separately run transfer experiments, so those point estimates need not coincide.

| Method            | PWH26  | PWH27  | PWH28  |
|-------------------|--------|--------|--------|
| Stay-still        | 1.8221 | 1.5355 | 1.0704 |
| Shared CFM        | 0.7110 | 0.2881 | 0.1850 |
| Mean displacement | 0.5674 | 0.3496 | 0.3388 |
| Nearest donor     | 0.6792 | 0.4318 | 0.4331 |

Table S5 (continued)

| Method                | PWH26  | PWH27  | PWH28  |
|-----------------------|--------|--------|--------|
| Conditioned CFM       | 1.4730 | 0.1990 | 0.1809 |
| Entropic OT extension | 0.7201 | 0.6228 | 0.8423 |

**Table S6.** Seed-averaged donor-count comparison for shared flow matching. ED is empirical energy distance and k is the number of training donors. At k=1, the two single-training-donor choices are averaged within each held donor and seed; each row then averages seeds 0, 1 and 2. Reduction is k=1 ED minus k=2 ED. The equally weighted mean reduction is 0.1655, with a 4,000-draw percentile 95% interval [0.0813, 0.3146] from three held-donor blocks. Training sets overlap; this descriptive interval is not a population coverage guarantee or evidence of a learning-curve plateau.

| Held donor | k=1 ED | k=2 ED | Reduction |
|------------|--------|--------|-----------|
| PWH26      | 1.0732 | 0.7586 | 0.3146    |
| PWH27      | 0.3373 | 0.2368 | 0.1005    |
| PWH28      | 0.2807 | 0.1994 | 0.0813    |

**Table S7.** Mouse POD7 audit for the source-labelled NDB, PDB and GDB arms. Each field uses rank time and POD0, POD2 and POD30 for fitting/scaling, then predicts from POD2 to excluded POD7. ED is empirical energy distance; improvement is 100 times (stay-still ED minus predicted ED) divided by stay-still ED. Each arm covers 4,182/7,002 human-panel genes and has one animal per time point. The ordering check asks whether endpoint distance is at least both endpoint-to-target distances; it is not proof of a geodesic. Values have no confidence intervals, and cell counts do not provide animal replication.

| Arm | Stay-still ED | Predicted ED | Improvement | Ordering check |
|-----|---------------|--------------|-------------|----------------|
| NDB | 1.2341        | 1.2525       | -1.49%      | Yes            |
| PDB | 2.4345        | 1.6526       | 32.12%      | No             |
| GDB | 1.3532        | 0.6682       | 50.62%      | Yes            |

**Table S8.** Fibroblast cell counts and target-sampling reference for each mouse arm. POD means postoperative day. POD0, POD2 and POD30 enter training; POD7 is the held-out target. Every arm/timepoint cell group comes from one animal, so these counts quantify distribution sampling and not biological replication. The final column is one seeded split-half POD7 energy distance in the training-standardized space. NDB, PDB and GDB are retained source-study arm labels. All arms share 4,182/7,002 mapped panel genes.

| Arm | POD0 | POD2 | POD7 held out | POD30 | Split-half ED |
|-----|------|------|---------------|-------|---------------|
| NDB | 2818 | 8177 | 12914         | 550   | 0.00544       |
| PDB | 1197 | 471  | 2500          | 813   | 0.01091       |
| GDB | 223  | 2400 | 11590         | 1402  | 0.00617       |

**Table S9.** Descriptive geometry for three healthy acute-wound donors. The first estimate divides the median of 18 within-donor between-time energy distances by the median of 12 between-donor same-time distances. The second is the mean of three pairwise cosines between donor Wound1-to-Wound7 mean-displacement vectors. Endpoints are percentile 95% limits from 10,000 resamples of pairwise distances within class or of cosine pairs. Pairs share donors, so these intervals do not provide independent donor replication or a general population ordering.

| Summary                    | Estimate | Lower endpoint | Upper endpoint |
|----------------------------|----------|----------------|----------------|
| time to donor median ratio | 8.4570   | 6.0173         | 13.1114        |
| mean displacement cosine   | 0.9477   | 0.9432         | 0.9536         |

**Table S10.** All six training-donor comparisons for shared and donor-conditioned flow matching. The first column identifies the donor excluded from that fit; the second is a retained training donor whose Wound1-to-Wound7 prediction is evaluated. ED is empirical energy distance in the fold-standardized coordinates. Both source and target of the evaluated donor were available during fitting, so these are fitting-capacity comparisons rather than donor-transfer tests. Conditioned errors are lower in all six rows. The repeated use of three donors prevents interpreting the rows as six independent biological replicates.

| Excluded donor | Evaluated training donor | Stay-still ED | Shared ED | Conditioned ED |
|----------------|--------------------------|---------------|-----------|----------------|
| PWH26          | PWH27                    | 1.6257        | 0.1277    | 0.0369         |
| PWH26          | PWH28                    | 1.0837        | 0.0930    | 0.0305         |
| PWH27          | PWH26                    | 1.7375        | 0.2483    | 0.0511         |
| PWH27          | PWH28                    | 1.0227        | 0.1248    | 0.0286         |
| PWH28          | PWH26                    | 1.7873        | 0.3089    | 0.0584         |
| PWH28          | PWH27                    | 1.6191        | 0.1190    | 0.0306         |

**Table S11.** Donor-count means by training seed and non-neural control. CFM denotes conditional flow matching, ED empirical energy distance and k the number of training donors. At k=1, two available single-donor fits are averaged within held donor; at k=2 one fit uses both available donors. Means then equally weight the three held donors. Reduction is k=1 minus k=2. CFM evaluation subsamples are fixed across seeds 0, 1 and 2; the mean-displacement control is recorded at seed 0. All nine CFM runs per seed and all six/three control runs at k=1/k=2 meet the descriptive split-half criterion. The control also improves with a second donor.

| Predictor         | Seed | k=1 mean ED | k=2 mean ED | Reduction |
|-------------------|------|-------------|-------------|-----------|
| Shared CFM        | 0    | 0.5496      | 0.3854      | 0.1643    |
| Shared CFM        | 1    | 0.6037      | 0.4276      | 0.1761    |
| Shared CFM        | 2    | 0.5380      | 0.3819      | 0.1560    |
| Mean displacement | 0    | 0.5325      | 0.4186      | 0.1139    |

**Table S12.** Training stochasticity for every held-donor/training-subset configuration. Mean ED, sample SD and range summarize the same fit repeated at seeds 0, 1 and 2; range is maximum minus minimum. Evaluation subsampling and the target split are fixed, so variation reflects training randomness. Averaging the nine configuration SDs gives 0.0717, 3.13 times the mean split-half reference 0.0229; the maximum range is 0.3683. Seeds and overlapping subsets are not independent donors and the single split-half criterion omits this training variability.

| Held donor | Training donors | k | Mean ED | SD     | Range  |
|------------|-----------------|---|---------|--------|--------|
| PWH26      | PWH27           | 1 | 1.3278  | 0.2070 | 0.3683 |
| PWH26      | PWH28           | 1 | 0.8186  | 0.0066 | 0.0125 |
| PWH26      | PWH27, PWH28    | 2 | 0.7586  | 0.1238 | 0.2334 |
| PWH27      | PWH26           | 1 | 0.2038  | 0.0291 | 0.0581 |
| PWH27      | PWH28           | 1 | 0.4709  | 0.1043 | 0.1896 |
| PWH27      | PWH26, PWH28    | 2 | 0.2368  | 0.0516 | 0.1021 |
| PWH28      | PWH27           | 1 | 0.3322  | 0.0878 | 0.1757 |
| PWH28      | PWH26           | 1 | 0.2293  | 0.0325 | 0.0605 |
| PWH28      | PWH26, PWH27    | 2 | 0.1994  | 0.0029 | 0.0057 |

**Table S13.** Fixed-field Runge–Kutta resolution comparison for pooled day-7 holdout. A single seed-0 field was refitted with the specified 8,000 optimization steps and 256-cell batches on the saved expression coordinates, excluding day 7 from fitting and scaling. All rows use the same 1,012 day-1 cells and the same 1,500 sampled target cells. Energy distance is the empirical V-statistic evaluated in double precision; improvement is relative to the same unchanged-source baseline (1.406924). RMS is the root-mean-square coordinate difference from the 200-step solution; it is a numerical discrepancy, not a confidence interval. The comparison is conditional on one field and finite-precision arithmetic and does not establish formal convergence to an exact ODE solution.

| RK4 steps | Energy distance | Improvement | RMS vs 200 steps |
|-----------|-----------------|-------------|------------------|
| 25        | 0.705395409     | 49.862585%  | 5.46e-07         |
| 50        | 0.705395449     | 49.862582%  | 4.76e-07         |
| 100       | 0.705395463     | 49.862581%  | 5.36e-07         |
| 200       | 0.705395452     | 49.862582%  | 0                |

**Table S14.** Evaluation-sampling sensitivity for the same fixed field with 50 RK4 steps. For each cell count, 50 seeded draws sample source and target cells without replacement; prediction and unchanged-source comparisons use identical source indices and identical target indices within a draw. An additional pair of disjoint target subsets at that count defines a split-half reference. Percentages summarize relative improvement over unchanged source; the central 95% draw range measures Monte Carlo sensitivity conditional on this cohort and field, not patient-level confidence. The final column counts draws whose gain exceeds the split-half reference. Cell count and resampling do not increase the three-donor biological sample size.

| Cells per distribution | Draws | Median improvement | Central 95% draw range | Gain exceeds reference |
|------------------------|-------|--------------------|------------------------|------------------------|
| 250                    | 50    | 48.46%             | 44.24 to 51.57%        | 50/50                  |
| 500                    | 50    | 48.39%             | 46.22 to 51.69%        | 50/50                  |
| 1000                   | 50    | 49.26%             | 47.86 to 50.60%        | 50/50                  |

**Table S15.** Evaluation of the saved pooled 50-step field in training-standardized coordinates, using all 1,012 Wound1 and 3,821 Wound7 cells. ED is the double-precision weighted empirical energy distance, without subsampling. Cell-pooled weights are uniform; donor-balanced weights allocate one third to each donor and equal mass within donor. Predictions retain source weights. The final three rows evaluate training donors separately, not donor-transfer folds. Improvement uses the corresponding unchanged-source ED. These three-donor point estimates have no confidence intervals and differ from capped-distance estimates.

| Evaluation       | Source / target cells | Stay-still ED | Predicted ED | Improvement |
|------------------|-----------------------|---------------|--------------|-------------|
| Cell-pooled      | 1012 / 3821           | 1.36146       | 0.68890      | 49.40%      |
| Equal donor mass | 1012 / 3821           | 1.31723       | 0.66265      | 49.69%      |
| PWH26            | 255 / 920             | 1.75803       | 0.92909      | 47.15%      |
| PWH27            | 497 / 1992            | 1.58457       | 0.84112      | 46.92%      |
| PWH28            | 260 / 909             | 1.07483       | 0.64420      | 40.07%      |

**Table S16.** Observed author-annotated fibroblasts retained after the sample–barcode join in GSE241132. Donors A, B and C correspond to source donor labels PWH26, PWH27 and PWH28. Each cell-count entry comes from one biopsy; the three donors supply the biological replication. The 12 entries sum to 12,259 cells and provide the material for expression and geometry displays. Day 0 is intact skin, followed by acute wounds at days 1, 7 and 30.

| Donor   | Day 0 cells | Day 1 cells | Day 7 cells | Day 30 cells |
|---------|-------------|-------------|-------------|--------------|
| Donor A | 724         | 255         | 920         | 843          |
| Donor B | 521         | 497         | 1992        | 2096         |
| Donor C | 907         | 260         | 909         | 2335         |

**Table S17.** Retrospective global missing-time comparison in three donors. Each error is the equally weighted mean of three within-donor, full-cell empirical energy distances in all 15 training-standardized coordinates. The table gives its mean and range over seeds 0, 1 and 2; deterministic baselines have zero seed range. Improvement is relative to the corresponding unchanged-source error. Withheld day 7 uses day 1 as source and day 30 as anchor; withheld day 1 uses intact skin as source and day 7 as anchor. Both have rank-time interpolation fraction one half. The held-time cells are excluded from scaling, fitting and anchor construction. Seed ranges are not donor confidence intervals, and absolute errors from different tasks use different scalers.

| Held day | Predictor            | Mean ED | Minimum ED | Maximum ED | Mean improvement |
|----------|----------------------|---------|------------|------------|------------------|
| 7        | Unchanged source     | 1.4725  | 1.4725     | 1.4725     | 0.00%            |
| 7        | Centroid translation | 0.6674  | 0.6674     | 0.6674     | 54.67%           |
| 7        | Independent bridge   | 0.9284  | 0.9224     | 0.9386     | 36.95%           |
| 7        | Shared flow          | 0.8110  | 0.8044     | 0.8237     | 44.93%           |
| 1        | Unchanged source     | 1.3460  | 1.3460     | 1.3460     | 0.00%            |
| 1        | Centroid translation | 0.9293  | 0.9293     | 0.9293     | 30.96%           |
| 1        | Independent bridge   | 1.0959  | 1.0920     | 1.1007     | 18.58%           |
| 1        | Shared flow          | 1.0571  | 0.9484     | 1.1231     | 21.46%           |

**Table S18.** Every donor-specific error in the two global missing-time tasks, summarized over the three computational seeds. ED is the full 15-dimensional empirical energy distance using all available source-derived predictions and target cells in that donor. The mean, minimum and maximum describe seeds, not independent specimens. The centroid baseline has lower seed-averaged error than the field in all three day-7 donors and two of three day-1 donors. Each donor supplies both training endpoints in this retrospective design; these rows are not held-out-donor transfer.

| Held day | Donor   | Predictor            | Mean ED | Minimum ED | Maximum ED |
|----------|---------|----------------------|---------|------------|------------|
| 7        | Donor A | Unchanged source     | 1.7580  | 1.7580     | 1.7580     |
| 7        | Donor A | Centroid translation | 0.7058  | 0.7058     | 0.7058     |
| 7        | Donor A | Independent bridge   | 0.8633  | 0.8508     | 0.8712     |
| 7        | Donor A | Shared flow          | 0.9069  | 0.8878     | 0.9291     |
| 7        | Donor B | Unchanged source     | 1.5846  | 1.5846     | 1.5846     |
| 7        | Donor B | Centroid translation | 0.6898  | 0.6898     | 0.6898     |
| 7        | Donor B | Independent bridge   | 0.9808  | 0.9591     | 0.9980     |
| 7        | Donor B | Shared flow          | 0.8645  | 0.8411     | 0.8804     |
| 7        | Donor C | Unchanged source     | 1.0748  | 1.0748     | 1.0748     |
| 7        | Donor C | Centroid translation | 0.6066  | 0.6066     | 0.6066     |
| 7        | Donor C | Independent bridge   | 0.9411  | 0.8981     | 0.9889     |
| 7        | Donor C | Shared flow          | 0.6614  | 0.6442     | 0.6868     |
| 1        | Donor A | Unchanged source     | 1.1341  | 1.1341     | 1.1341     |
| 1        | Donor A | Centroid translation | 0.8704  | 0.8704     | 0.8704     |
| 1        | Donor A | Independent bridge   | 1.0311  | 1.0192     | 1.0471     |
| 1        | Donor A | Shared flow          | 0.8410  | 0.7370     | 0.8960     |
| 1        | Donor B | Unchanged source     | 1.5517  | 1.5517     | 1.5517     |
| 1        | Donor B | Centroid translation | 1.0900  | 1.0900     | 1.0900     |
| 1        | Donor B | Independent bridge   | 1.3196  | 1.3132     | 1.3232     |
| 1        | Donor B | Shared flow          | 1.3462  | 1.2486     | 1.4146     |
| 1        | Donor C | Unchanged source     | 1.3522  | 1.3522     | 1.3522     |
| 1        | Donor C | Centroid translation | 0.8275  | 0.8275     | 0.8275     |
| 1        | Donor C | Independent bridge   | 0.9369  | 0.9258     | 0.9525     |
| 1        | Donor C | Shared flow          | 0.9841  | 0.8597     | 1.0585     |